Gene Essentiality from the Cell Graph

The backbone against flux balance analysis and Flux Cone Learning,

and why the solve helps the prediction where the metabolic module does not

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Abstract

Gene essentiality is the one endpoint on which a flux model has a published, non-trivial score: on the released Merzbacher 2025 yeast split, flux balance analysis reads an AUROC of 0.69 over all 1,121 labeled genes and Flux Cone Learning 0.73 on the 223 held-out genes. This document scores the cell graph transformer on the same held-out genes, puts FBA on every medium the ontology defines and a classical ladder of models on the same split, and asks two questions. The first is how far the held-out AUROC goes: a width-12 transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, scores 0.893 [0.84, 0.94] under Flux Cone Learning's own fold protocol, 0.15 above it on the same genes, and every part of the gain is an input the ontology computes for every gene at no training cost. The second is why the solve helps and the metabolic module, which is built on the same genome-scale model, does not: routed through the module's growth column the prediction never beats the backbone alone, and the rounds locate the reason in the layer, whose alternating projections never reach the polytope FBA optimizes over and which discards the solve when handed it. Every number here is at a validation-selected epoch on a pinned test set, with a gene bootstrap and the across-seed range, and every table is generated from the result files.

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Contents

1 The endpoint, its labels and its splits ✗ . . . **3**
1.1 Vocabulary ✗ . . . 3
1.2 Where the labels come from ✗ . . . 3
1.3 The protocol inherited from 027, and what it got wrong ✗ . . . 4
2 The comparators, computed rather than quoted ✗ . . . **4**
2.1 Flux balance analysis on four media ✗ . . . 4
2.2 yeast-GEM's own benchmark, and accuracy against balanced accuracy ✗ . . . 4
2.3 Flux Cone Learning on the same genes ✗ . . . 5
2.4 The classical ladder on the same split ✗ . . . 5
2.5 Graph-only baselines: message passing over the gene graph ✗ . . . 7
3 The backbone: splits, labels, weight decay and the shape of the curve ✗ . . . **8**
3.1 The curve was noisy because the validation set was small ✗ . . . 8
3.2 The split file costs training genes, and the cells are under-trained ✗ . . . 10
3.3 Weight decay and graph regularization do nothing here ✗ . . . 10
3.4 More labels from the genome make the held-out ranking worse ✗ . . . 10
3.5 Gradient norms ✗ . . . 10
3.6 Round 3: the two levers combined, a longer budget, and the fold ensemble ✗ . . . 10
3.7 Round 5: FBA's growth ratios as an input ✗ . . . 12
3.8 Round 8: width and depth on the FBA cell ✗ . . . 12
3.9 Round 9: the FBA cell on the ladder's other encodings ✗ . . . 15
3.10 Round 12: the passing cell confirmed, and the number to report ✗ . . . 16
3.11 Round 11: the affine widened with the solve ✗ . . . 16
3.12 Round 14: which columns of the indicator carry the gain ✗ . . . 19
3.13 Round 16: the round-9 backbone with the round-11 affine ✗ . . . 20
3.14 Round 17: the round-16 cell on the five folds ✗ . . . 21
4 The metabolic module: the solve helps, so why not the layer? ✗ . . . **23**
4.1 What the layer's flux violates ✗ . . . 23
4.2 Round 2: the module on the stable backbone, with and without the penalties ✗ . . . 24
4.3 Round 4: the module on the best backbone ✗ . . . 26
4.4 Round 6: one medium against four, the layer on top of the solve, and two other embeddings ✗ . . . 26
4.5 The growth column is a signed free coordinate ✗ . . . 29

4.6 Round 7: the growth column with the bounds enforced ✗ . . . 30
4.7 Round 10: the layer warm-started from the deletion's own solve ✗ . . . 30
4.8 Round 13: the warm start on the stoichiometry-only layer ✗ . . . 33
4.9 Round 15: the layer that solves ✗ . . . 33
4.10 Where the gradient goes ✗ . . . 36
4.11 Why the solve helps and the layer does not ✗ . . . 36
5 Open ✗ . . . 37

1 The endpoint, its labels and its splits ✗

1.1 Vocabulary ✗

AUROC Area under the receiver operating characteristic, computed as the Mann-Whitney rank statistic with mid-ranks on ties: the probability that a randomly drawn essential gene scores above a randomly drawn nonessential one.

backbone The cell graph transformer over the nine-graph perturbed gene graph with `prot_T5_all` node features unless another encoding is named, read at the pooled perturbed tokens. The *pooled* arm is this backbone with a scalar head and no metabolic module.

chromosome-pathway indicator The 197-dimensional per-gene encoding `normalized_chrom_pathways` (`sgd_gene_graph_hot.py`): five SGD numeric gene features scaled to $[0, 1]$ (protein length, molecular weight, isoelectric point, the median and the median absolute deviation of the gene's expression), a one-hot of the gene's chromosome, and a multi-hot over the SGD pathways it is annotated to. *chrom-pathway* in figure labels; the name understates it, since five of its columns are continuous gene properties and two of those are expression.

FBA affine The readout that adds $w^\top r + b$ to the backbone's logit, where r holds the deleted gene's knockout-to-wild-type growth ratio on each of four media from one FBA solve per gene; the *fba* arm.

fold ensemble Five models trained on the five cross-validation files (each a different fifth of the 787 labeled non-test genes held for validation), their logits averaged on the held-out genes; Flux Cone Learning's own test protocol.

gene bootstrap Resampling the scored genes with replacement (2,000 draws) and reporting the 2.5 and 97.5 percentiles of the statistic.

held-out genes The 223 genes Merzbacher 2025 held out of its own training (`test = 1` in the released split), of which 195 resolve as single-deletion genotypes in the build. Every model number in this document is on those 195 unless the gene set is named.

metabolic module The enzyme-constrained thermodynamic flux layer over yeast-GEM 9.0.2 that maps the perturbed representation to a flux vector by K alternating projections onto the mass balance $Sv = 0$ and the box of bounds; here it is read at the objective column `r_2111` (growth). The *anchored* layer tightens the GEM's box by the enzyme-capacity ceilings and the second-law directions; the *stoichiometry-only* layer keeps the GEM's box.

parsimonious FBA Two linear programs: the growth optimum over the polytope $\{Sv = 0,\ \ell \le v \le u\}$ with the deleted gene's reactions closed through the gene-protein-reaction rules, then the flux of least total absolute value at that optimum. *pFBA*; *the solve* when the object is the optimum rather than the method.

released labels The 1,121 essentiality calls in the released split. The column named `essential` is a viability flag (1.0 marks a nonessential gene); it is inverted once and the four class counts are asserted against the paper's Methods.

selected epoch The argmax of a centered five-epoch moving average of the validation AUROC. The held-out genes are scored once, at that epoch.

solve-derived encodings Per-gene vectors computed from pFBA rather than from sequence: the deletion's own flux vector, its difference from the wild-type vector, and eight statistics of the gene's reactions under the wild-type solve and its flux variability (`ess_reference_flux_table.py`).

warm start The flux layer's proposal written as a reference flux plus a null-space residual of relative size α, $v = v_{\text{ref}} + \alpha N z$, before the alternations; with a per-genotype table the reference is the deletion's own pFBA flux.

1.2 Where the labels come from ✗

The build is a union of the SGD essential-gene list (1,140 genotypes, all essential) and the single-mutant fitness screens (Costanzo 2016, Kuzmin 2018; 4,540 genotypes), 5,656 single-deletion genotypes in all. No fitness value enters any objective: the head is one logit scored by binary cross entropy against a 0/1 label, and every genotype whose gene carries no label arrives masked. Two label sources are used and named throughout:

Table 1: The split files, one per label source: genes per split with the essential count in parentheses. The test block is the released held-out genes under either source; validation is a class-stratified fifth of the remaining labeled genes at seed 0. Under the genome source the released label wins on every disagreement (12 genes).

split file	labels	train (ess)	val (ess)	test (ess)
`ess_fcl_val0.2_s0`	released (1,121)	629 (100)	158 (25)	195 (31)
`ess_genome_val0.2_s0`	released + SGD essential + viable fitness (5,795)	4,368 (879)	1,093 (220)	195 (31)

1.3 The protocol inherited from 027, and what it got wrong ✗

The wave-9 protocol pinned the held-out genes and let the data module draw train and validation at 80/10/10 over the remaining 5,461 records. Of those, 926 carry a label, so the validation set held 75 labeled genes with about 12 essential, on which a single AUROC read has a Hanley-McNeil standard error near 0.07; 10 percent of the labeled genes sat in a random test block nothing reports; and 4,535 unlabeled records were forwarded every epoch under a False mask. A split file replaces the draw: it names every record's split, is class-stratified, carries the labels it was stratified on, and the trainer drops every record it does not name.

2 The comparators, computed rather than quoted ✗

2.1 Flux balance analysis on four media ✗

The published FBA comparator is cobrapy `single_gene_deletion` with glucose as the carbon source, scored on all 1,121 labeled genes. The phenotype it is compared against was called on rich medium, and the model here is scored on the 195 held-out genes, so the comparison as published is neither on the same medium nor on the same genes. Table 2 recomputes it on yeast-GEM 9.0.2 under the exchange bounds the model ships with and under the three media the ontology defines, with the essentiality score being the negative knockout-to-wild-type growth ratio, rounded to 10^{-3} so that the roughly 900 deletions that grow at 0.99 to 1.00 of the wild type rank as the tie they are (without the rounding two identical runs differed by 0.03 AUROC on the held-out genes).

Table 2: FBA single-gene deletion on yeast-GEM 9.0.2 (cobrapy 0.30.0), scored as the negative knockout-to-wild-type growth ratio rounded to 10^{-3}. AUROC on all 1,121 labeled genes, on the 223 held-out genes, and on the 195 held-out genes the model scores with a 2,000-draw gene bootstrap interval. 1121 of the labeled genes are in the GEM; the rest keep wild-type growth. Bold marks the highest value in a column.

medium	WT growth (1/h)	lethal	all 1,121	held-out 223	our 195 [95% CI]
shipped bounds (glucose minimal, aerobic)	0.086	186	0.707	0.633	0.654 [0.56, 0.75]
SM, synthetic minimal	0.314	160	0.707	0.642	0.666 [0.57, 0.76]
SC, synthetic complete	0.543	91	**0.719**	**0.653**	**0.666** [0.57, 0.76]
YPD approximation (no peptone)	0.535	104	0.715	**0.653**	0.664 [0.57, 0.76]

The medium moves the number of lethal deletions by a factor of two and the AUROC by about 0.01. The published 0.69 on all labeled genes is reproduced to within 0.02.

2.2 yeast-GEM's own benchmark, and accuracy against balanced accuracy ✗

yeast-GEM ships its own essentiality benchmark and prints it in its README: on 1,107 genes, an accuracy of 0.902 with 933 true non-essential, 65 true essential, 94 false non-essential and 15 false essential calls. Its release file (`results/external/yeast-GEM-9.1.1/`, version 9.1.1 at commit `2d594ae`, 2026-08-30) says how: the Kennedy synthetic complete medium is written onto the exchanges (every exchange closed, 28 uptakes at -0.5, glucose at -20, fifteen inorganic uptakes open), each gene is deleted through the gene-protein-reaction rules, and a deletion is inviable when its growth ratio is below 10^{-6}; the genes are the SGD verified ORFs in the model, and the labels are the Stanford deletion project's inviable list of 2011, scored against the model at that one rule, a protocol the repository attributes to Heavner and Price (2015). The released split's labels are the same truth: on the 1,104 genes both name, the two sources agree on 1,102.

Table 3 puts every scorer in this document at that rule. Two things are visible at once. The first is that accuracy is the wrong statistic for this endpoint: 86 percent of the genes are nonessential, so a scorer that calls nothing essential reads 0.856 on yeast-GEM's set and 0.841 on the 195 held-out genes, and yeast-GEM's 0.902 is that floor plus four and a half points; its recall on the essential class is 0.41 and its balanced accuracy 0.70. FBA on the previous release, 9.0.2, under the ontology's own synthetic complete recipe reads 0.907 on the released labels with a recall of 0.46, the same rule on the same truth with a different medium composition, so the 9.1.1 number is reproduced to within half a point. The second is where the transformer sits at a threshold: at its loss's decision boundary it reads 0.856, the majority floor, with a recall of 0.68 and a balanced accuracy of 0.78, and at the threshold that calls 31 genes essential it reads 0.867 with precision and recall both at 0.58. The released Flux Cone Learning predictions at $P(\text{viable}) < 0.5$ call eight genes essential on the 195, all eight correctly, for a recall of 0.26. A ranking statistic is what separates the scorers, and the balanced accuracies, 0.63 to 0.72 for the solvers and 0.78 for the transformer, order them the way the AUROCs do.

Table 3: Every scorer at the rule yeast-GEM reports its benchmark with: a deletion is called essential when the criterion in the rule column holds, and the call is scored against the label as a confusion matrix with essential as the positive class (TP/FN/FP/TN). Accuracy is followed by the majority class's share of the gene set in parentheses; precision and recall are for the essential class; balanced accuracy is the mean of recall and specificity. yeast-GEM 9.1.1's row is its own release file on its own labels (the Stanford deletion project's inviable list over the model's verified ORFs); those labels and the released split agree on 1102 of the 1,104 genes both name. Every other row is on the released labels. The transformer is the round-12 fold ensemble at the loss's decision boundary and at the threshold that calls as many genes essential as the labels hold. Bold marks the highest value in a column.

scorer	rule	gene set	ess/n	TP/FN/FP/TN	acc (maj)	prec	rec	spec	bal
yeast-GEM 9.1.1 (its own benchmark)	growth ratio < 1e-6, Kennedy SC medium	1,107 verified ORFs, Stanford inviable list	159/1,107	65/94/15/933	0.902 (0.856)	0.812	0.409	0.984	0.696
FBA, yeast-GEM 9.0.2, shipped	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	82/75/104/860	0.840 (0.860)	0.441	0.522	0.892	0.707
FBA, yeast-GEM 9.0.2, shipped	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/11/153	0.841 (0.841)	0.500	0.355	0.933	0.644
FBA, yeast-GEM 9.0.2, SM	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	78/79/82/882	0.856 (0.860)	0.487	0.497	0.915	0.706
FBA, yeast-GEM 9.0.2, SM	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/7/157	0.862 (0.841)	0.611	0.355	0.957	0.656
FBA, yeast-GEM 9.0.2, SC	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	72/85/19/945	**0.907** (0.860)	0.791	0.459	0.980	0.719
FBA, yeast-GEM 9.0.2, SC	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/2/162	0.887 (0.841)	0.846	0.355	0.988	0.671
FBA, yeast-GEM 9.0.2, YPD	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	73/84/31/933	0.897 (0.860)	0.702	0.465	0.968	0.716
FBA, yeast-GEM 9.0.2, YPD	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/2/162	0.887 (0.841)	0.846	0.355	0.988	0.671
Flux Cone Learning (released predictions)	P(viable) < 0.5	223 released held-out (released labels)	31/223	8/23/0/192	0.897 (0.861)	**1.000**	0.258	**1.000**	0.629
Flux Cone Learning (released predictions)	P(viable) < 0.5	195 held-out (released labels)	31/195	8/23/0/164	0.882 (0.841)	**1.000**	0.258	**1.000**	0.629
FBA (released predictions)	growth < 1e-6	195 held-out (released labels)	31/195	11/20/11/153	0.841 (0.841)	0.500	0.355	0.933	0.644
transformer, round-12 fold ensemble	logit > 0	195 held-out (released labels)	31/195	21/10/18/146	0.856 (0.841)	0.538	**0.677**	0.890	**0.784**
transformer, round-12 fold ensemble	top 31 logits (label prevalence)	195 held-out (released labels)	31/195	18/13/13/151	0.867 (0.841)	0.581	0.581	0.921	0.751

2.3 Flux Cone Learning on the same genes ✗

Recomputed from the authors' released per-gene predictions: 0.7308 on the 223 held-out genes (published 0.73) and 0.742 on the 195 genes the model scores. On those same 195 genes the released FBA predictions score 0.640, so the published pair overstates FBA by about 0.06 relative to a head-to-head on one gene set. The paper prints a third number, 0.78 ± 0.065 in the legend of its Figure 3a, which is the mean over five cross-validation folds drawn inside its 898 training genes; a model scored on the held-out genes is compared with the 0.73 line, never with that one.

2.4 The classical ladder on the same split ✗

The fitness and expression lines each carry a ladder of models that need no graph and no training beyond a solve, and essentiality gets the classification form of it: FBA's growth ratio with no fitting, then four model classes (L_2 logistic regression, a random forest, a radial-basis support vector classifier, cosine nearest neighbors) on each of twenty-four gene encodings, every pair fit twice, with and without the four FBA growth ratios appended. The encodings are the seventeen the fitness ladder carries (codon frequency, the chromosome and pathway indicator, six nucleotide windows, the two FUDT embeddings, the species language model, CaLM, ProtT5 and ESM2 with and without dubious genes, and their concatenation, `mixed`) plus seven controls: a one-hot gene identity, and standard normal vectors of dimension 1, 10, 100, 1,000, 6,607 (the one-hot's width) and 8,192, drawn once at seed 0. A one-hot rung must score chance on held-out genes, since every held-out column is zero in training, and a random vector must score chance at every width; either scoring above chance would mean the split leaks. Every fitted rung is fit on the split file's 629 training genes, its one hyperparameter chosen by five-fold stratified cross-validation inside those genes, and scored once on the 158 validation and 195 held-out genes, on which nothing was chosen. The transformer cells in Sections 3 and 4 train on the same 629 genes and score on the same 195. Three further encodings come from the solve itself rather than from the sequence, added after round 8 (`ess_reference_flux_table.py`): the deletion's own parsimonious FBA flux vector on YPD (4,131 reactions), that vector's difference from the wild-type vector, and eight numbers about the gene's own reactions under the wild-type solve and its flux variability at 90 percent of the optimum (reaction count, reactions carrying flux, the largest and the summed absolute flux, the largest forced flux, the smallest and largest variability range, and the deletion's growth ratio). Figure 1 shows all 216 fitted rungs; Table 4 keeps, per encoding, the model class whose cross-validation AUROC is highest.

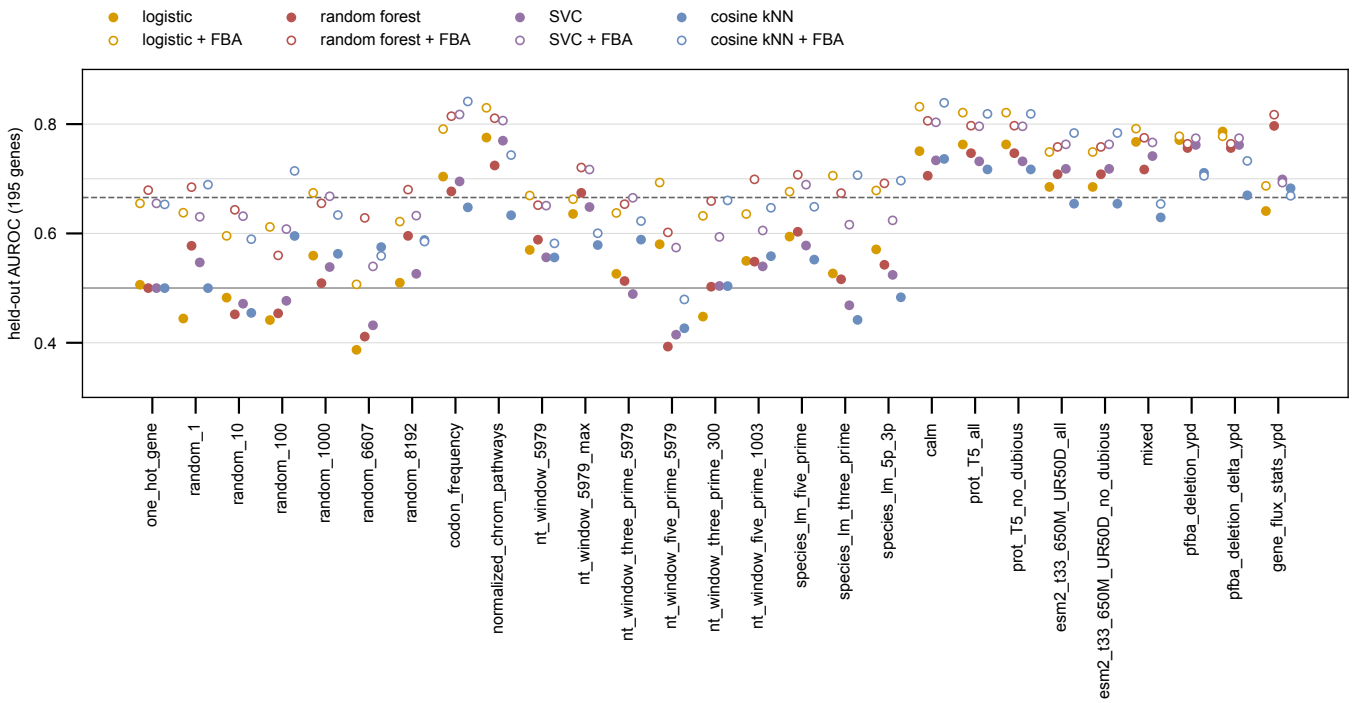


Figure 1: The classical ladder on the released-label split. Held-out AUROC on the 195 genes for every model class (color) on every encoding (column), filled without and open with the four FBA growth ratios appended; the dashed line is FBA alone on the shipped medium (0.654), the solid line chance. The last three columns are the solve-derived encodings. The hyperparameter of each mark was chosen by five-fold cross-validation inside the 629 training genes. Written by `experiments/028-gene-essentiality/scripts/logistic_baselines.py`.

Table 4: The classical ladder on the released-label split (`ess_fcl_val0.2_s0.json.gz`). Per encoding, the model class whose five-fold cross-validation AUROC inside the 629 training genes is highest, with and without FBA's four growth ratios appended; the validation genes and the 195 held-out genes (31 essential) are scored once, on genes no fit touched, the held-out number with a 2,000-draw gene bootstrap. FBA alone is not fitted. Every model and encoding is in Figure 1. Bold marks the highest value in a column.

encoding	best model (chosen)	dim	CV	val	held-out [95% CI]
FBA growth	medium=shipped	–	–	0.683	0.654 [0.56, 0.75]
FBA growth	medium=SM	–	–	0.710	0.666 [0.57, 0.76]
FBA growth	medium=SC	–	–	0.726	0.666 [0.57, 0.76]
FBA growth	medium=YPD	–	–	0.714	0.664 [0.57, 0.76]
one_hot_gene	logistic (C=10)	6,579	0.504	0.508	0.506 [0.50, 0.52]
one_hot_gene + FBA	random forest (max_depth=8)	6,583	0.775	0.730	0.679 [0.59, 0.77]
random_1	random forest (max_depth=4)	1	0.598	0.560	0.577 [0.46, 0.69]
random_1 + FBA	cosine kNN (k=10)	5	0.806	0.699	0.689 [0.56, 0.80]
random_10	cosine kNN (k=50)	10	0.549	0.481	0.455 [0.35, 0.56]
random_10 + FBA	logistic (C=0.001)	14	0.786	0.684	0.596 [0.46, 0.72]
random_100	random forest (max_depth=None)	100	0.522	0.376	0.454 [0.34, 0.57]
random_100 + FBA	random forest (max_depth=4)	104	0.773	0.662	0.560 [0.42, 0.70]
random_1000	random forest (max_depth=4)	1,000	0.475	0.490	0.509 [0.40, 0.62]
random_1000 + FBA	random forest (max_depth=None)	1,004	0.778	0.749	0.655 [0.53, 0.78]
random_6607	cosine kNN (k=1)	6,607	0.549	0.477	0.575 [0.49, 0.66]
random_6607 + FBA	random forest (max_depth=None)	6,611	0.734	0.713	0.628 [0.50, 0.75]
random_8192	cosine kNN (k=50)	8,192	0.523	0.515	0.588 [0.49, 0.69]
random_8192 + FBA	random forest (max_depth=8)	8,196	0.768	0.748	0.680 [0.57, 0.79]
codon_frequency	SVC (C=1)	64	0.700	0.663	0.695 [0.59, 0.79]
codon_frequency + FBA	SVC (C=1)	68	0.849	0.772	0.818 [0.73, 0.90]
normalized_chrom_pathways	logistic (C=10)	197	0.705	0.722	0.775 [0.68, 0.86]
normalized_chrom_pathways + FBA	logistic (C=0.01)	201	0.802	0.791	**0.830** [0.75, 0.90]
nt_window_5979	logistic (C=0.01)	2,560	0.579	0.532	0.570 [0.46, 0.67]
nt_window_5979 + FBA	logistic (C=0.01)	2,564	0.743	0.705	0.669 [0.55, 0.79]
nt_window_5979_max	logistic (C=0.001)	2,560	0.626	0.608	0.636 [0.54, 0.73]
nt_window_5979_max + FBA	logistic (C=0.01)	2,564	0.777	0.786	0.663 [0.54, 0.78]
nt_window_three_prime_5979	logistic (C=0.001)	2,560	0.555	0.580	0.526 [0.41, 0.64]
nt_window_three_prime_5979 + FBA	logistic (C=0.001)	2,564	0.732	0.720	0.637 [0.51, 0.76]
nt_window_five_prime_5979	cosine kNN (k=25)	2,560	0.594	0.518	0.427 [0.32, 0.54]
nt_window_five_prime_5979 + FBA	logistic (C=0.01)	2,564	0.733	0.657	0.693 [0.59, 0.79]
nt_window_three_prime_300	random forest (max_depth=4)	2,560	0.564	0.537	0.503 [0.39, 0.62]

encoding	best model (chosen)	dim	CV	val	held-out [95% CI]
nt_window_three_prime_300 + FBA	random forest (max_depth=None)	2,564	0.752	0.757	0.659 [0.53, 0.78]
nt_window_five_prime_1003	logistic (C=0.001)	2,560	0.541	0.485	0.550 [0.43, 0.68]
nt_window_five_prime_1003 + FBA	logistic (C=0.01)	2,564	0.727	0.665	0.636 [0.52, 0.75]
fudt_upstream	SVC (C=1)	768	0.611	0.583	0.578 [0.47, 0.69]
fudt_upstream + FBA	random forest (max_depth=None)	772	0.775	0.747	0.707 [0.60, 0.81]
fudt_downstream	SVC (C=10)	768	0.589	0.553	0.469 [0.36, 0.58]
fudt_downstream + FBA	logistic (C=0.1)	772	0.770	0.735	0.706 [0.59, 0.81]
species_lm_5p_3p	logistic (C=0.001)	1,536	0.634	0.631	0.571 [0.47, 0.68]
species_lm_5p_3p + FBA	logistic (C=0.01)	1,540	0.763	0.733	0.679 [0.56, 0.79]
calm	SVC (C=1)	768	0.755	0.777	0.734 [0.64, 0.82]
calm + FBA	random forest (max_depth=8)	772	**0.850**	0.833	0.806 [0.72, 0.89]
prot_T5_all	SVC (C=10)	1,024	0.781	0.767	0.732 [0.62, 0.83]
prot_T5_all + FBA	logistic (C=0.01)	1,028	0.842	0.826	0.821 [0.74, 0.90]
prot_T5_no_dubious	SVC (C=10)	1,024	0.781	0.767	0.732 [0.62, 0.83]
prot_T5_no_dubious + FBA	logistic (C=0.01)	1,028	0.842	0.826	0.821 [0.74, 0.90]
esm2_t33_650M_UR50D_all	random forest (max_depth=8)	1,280	0.731	0.694	0.708 [0.61, 0.80]
esm2_t33_650M_UR50D_all + FBA	random forest (max_depth=8)	1,284	0.815	0.794	0.758 [0.66, 0.86]
esm2_t33_650M_UR50D_no_dubious	random forest (max_depth=8)	1,280	0.731	0.694	0.708 [0.61, 0.80]
esm2_t33_650M_UR50D_no_dubious + FBA	random forest (max_depth=8)	1,284	0.815	0.794	0.758 [0.66, 0.86]
mixed	logistic (C=10)	6,149	0.788	0.794	0.768 [0.66, 0.87]
mixed + FBA	logistic (C=10)	6,153	0.833	**0.837**	0.792 [0.69, 0.89]
pfba_deletion_ypd	random forest (max_depth=None)	4,131	0.770	0.723	0.756 [0.65, 0.86]
pfba_deletion_ypd + FBA	random forest (max_depth=None)	4,135	0.776	0.736	0.764 [0.66, 0.86]
pfba_deletion_delta_ypd	random forest (max_depth=None)	4,131	0.769	0.723	0.756 [0.65, 0.86]
pfba_deletion_delta_ypd + FBA	random forest (max_depth=None)	4,135	0.776	0.736	0.764 [0.66, 0.86]
gene_flux_stats_ypd	random forest (max_depth=None)	8	0.837	0.760	0.797 [0.70, 0.88]
gene_flux_stats_ypd + FBA	random forest (max_depth=None)	12	0.847	0.777	0.817 [0.73, 0.90]

The controls behave. The one-hot rung scores 0.506 and the six random widths score 0.45 to 0.59, chance within the 195-gene noise, and with FBA appended they score 0.56 to 0.69, FBA's own number, so no rung reads anything but its features. Without FBA the protein and codon language models are the only encodings clear of 0.70: ProtT5 0.732, CaLM 0.734, the concatenation 0.768, and the chromosome-pathway indicator 0.775 [0.68, 0.86], each level with Flux Cone Learning's 0.742 on the same genes. FBA lifts every one of them: logistic regression on ProtT5 with the ratios scores 0.821 [0.74, 0.90] from 1,029 parameters, the support vector classifier on 64 codon frequencies 0.818 [0.73, 0.90], the random forest on CaLM 0.806 [0.72, 0.89], and logistic regression on the 197-dimensional chromosome-pathway indicator 0.830 [0.75, 0.90]. The rung with the highest cross-validation AUROC, the random forest on CaLM with FBA at 0.850, scores 0.806 on the held-out genes; the highest held-out number in the figure, 0.841 for cosine nearest neighbors on codon frequency with FBA, is the maximum over 192 rungs and is not a selected model. Neither an embedding nor the solver reaches 0.80 alone, and their sum does on four encodings. FBA's ratio is a feature the ontology computes for every gene at no training cost, so the transformer is measured against 0.80 to 0.83 from here on, not against the published 0.73.

The solve carries more than its growth ratio. The eight reaction statistics under a random forest score 0.797 [0.70, 0.88] with no sequence information, and 0.817 [0.73, 0.90] with the four ratios appended, at the highest cross-validation AUROC in the ladder (0.847); the growth ratio alone, which is one of the eight, scores 0.645 on the same genes. The full deletion flux vector scores 0.756 under its selected model and 0.771 under logistic regression, level with ProtT5 and with the pooled transformer, so a gene's position in the wild-type flux solution and how much the solve moves when it is deleted rank essential genes about as well as its protein sequence does, and the two sources are different: no other encoding reaches 0.80 without the solve, and the eight statistics come within 0.003 of it with nothing else.

2.5 Graph-only baselines: message passing over the gene graph ✗

The Gene_Graph project (2021 to 2023) classified essential genes by two-layer graph convolution and graph attention over YeastMine's physical, genetic and regulatory interaction graphs with all-ones node inputs, and saved no number. The same design is rerun here on the released split (`graph_baselines.py`): transductive node classification on one gene graph, where every gene's logit is a function of its neighborhood and its own features and no genotype is perturbed, on the physical interaction graph and on the union of the nine graphs the transformer reads, with all ones, with log degree and the clustering coefficient, or with ProtT5 as inputs, beside an MLP on ProtT5 with no graph (Table 5).

Table 5: Graph-only baselines on the released split: transductive node classification on one gene graph with no genotype perturbed, two layers of width 64, class-weighted binary cross entropy, 200 epochs, the epoch selected by a five-epoch moving average of the validation AUROC, scored once on the 195 held-out genes at seeds 2700 to 2702. The graph is the physical interaction graph or the union of the nine gene graphs the transformer reads; the inputs are all ones (the Gene_Graph design), log degree with the clustering coefficient, or ProtT5. The MLP row has no graph. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
MLP	none	ProtT5	1,024	69,825	0.754	0.003	0.756, 0.755, 0.750	62, 58, 56
GCN	physical	ones	1	4,353	0.579	0.003	0.577, 0.577, 0.583	88, 100, 49
GCN	physical	degree + clustering	2	4,417	0.582	0.009	0.592, 0.571, 0.583	29, 26, 3
GCN	physical	ProtT5	1,024	69,825	0.769	0.002	0.770, 0.766, 0.771	194, 193, 197
GATv2	physical	ones	1	8,897	0.495	0.034	0.455, 0.493, 0.538	137, 148, 50
GATv2	physical	degree + clustering	2	9,025	0.592	0.003	0.592, 0.588, 0.596	160, 192, 116
GATv2	physical	ProtT5	1,024	139,841	0.760	0.018	0.747, 0.747, 0.786	101, 93, 79
GCN	union of nine	ones	1	4,353	0.613	0.000	0.613, 0.613, 0.613	8, 5, 3
GCN	union of nine	degree + clustering	2	4,417	0.638	0.021	0.661, 0.644, 0.610	197, 197, 2
GCN	union of nine	ProtT5	1,024	69,825	**0.834**	0.003	0.836, 0.829, 0.836	197, 196, 197
GATv2	union of nine	ones	1	8,897	0.560	0.014	0.578, 0.556, 0.545	38, 107, 125
GATv2	union of nine	degree + clustering	2	9,025	0.702	0.025	0.699, 0.672, 0.733	197, 191, 197
GATv2	union of nine	ProtT5	1,024	139,841	0.796	0.020	0.809, 0.768, 0.811	118, 197, 119

Structure alone carries little and structure with sequence carries a lot. With all-ones inputs the models read 0.50 to 0.61, chance to the ladder's random-vector level, and with degree and clustering 0.58 to 0.70. With ProtT5 the MLP reads 0.754, the ladder's logistic rung plus 0.02, and graph convolution over the union of the nine graphs reads 0.836, 0.829 and 0.836, a seed mean of 0.834 with a spread of 0.003 from 69,825 parameters: the edges add 0.08 to the protein embedding, the physical graph alone adds 0.015, and attention adds nothing over convolution on either graph. That cell is above the pooled transformer (0.774) and level with the transformer with FBA's four ratios (0.821) with no solve at all, and its three seeds select epochs 196 to 197 of 200 with the validation curve still rising, so the budget is what bounds it. The transformer reads the same nine graphs and the same embedding; what it adds over this cell, 0.87 with the pathway indicator and the ratios, is the perturbed-graph reading and the two inputs, and the graph-convolution cell with those two inputs is the untested comparison (Section 5).

3 The backbone: splits, labels, weight decay and the shape of the curve ✗

Every cell in this section is the pooled backbone at width 12 with one transformer block and a one-layer preprocessor (16,874 parameters), trained for 80 epochs at three initialization seeds with the split pinned. Round 1 (jobs 2149 to 2181) moves one lever at a time away from the cell that trains on the released-label split file (`ess_fcl_val0.2_s0`, 629 training genes, 158 validation genes): the old 75-gene validation draw, the genome-wide labels, weight decay at 0, 10^{-2} and 10^{-1}, batch 32, learning rate at 3×10^{-4} and 3×10^{-3}, the class weight 772/126, and graph regularization off. Table 6 carries every number; Figure 2 the curves.

Table 6: Round 1, the backbone. Width 12, one block, one-layer preprocessor, 80 epochs, seeds 2700 to 2702. Held-out AUROC on the 195 genes: the seed mean with its standard deviation, the three seeds, the gene bootstrap of the pooled per-gene logits, the epochs selected on validation, and the median absolute epoch-to-epoch step of the validation curve. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
wave-11 protocol (75-gene val)	0.759	0.020	0.781, 0.755, 0.741	0.760 [0.67, 0.84]	18, 38, 14	0.0056
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708 ↓	0.738 [0.65, 0.83]	66, 77, 77	0.0026
weight decay 0	0.725	0.019	0.745, 0.720, 0.709 ↓	0.735 [0.64, 0.82]	66, 77, 77	0.0027
weight decay 1e-2	0.726	0.017	0.745, 0.722, 0.711 ↓	0.735 [0.64, 0.83]	65, 77, 77	0.0029
weight decay 1e-1	0.725	0.018	0.746, 0.721, 0.709 ↓	0.735 [0.64, 0.82]	66, 77, 77	0.0025
batch 32	**0.765**	0.005	0.766, 0.770, 0.759	0.764 [0.67, 0.85]	74, 73, 77	0.0032
genome labels (5,795 genes)	0.704	0.024	0.706, 0.726, 0.679 ↓	0.705 [0.61, 0.80]	77, 77, 69	0.0020
lr 3e-4	0.658	0.093	0.737, 0.682, 0.556 ↓	0.732 [0.64, 0.82]	77, 77, 74	0.0023
lr 3e-3	0.686	0.043	0.718, 0.704, 0.637 ↓	0.713 [0.62, 0.81]	77, 77, 50	0.0024
class weight 772/126	0.759	0.016	0.764, 0.772, 0.741	**0.769** [0.68, 0.86]	77, 75, 53	0.0034
graph regularization off	0.725	0.019	0.745, 0.722, 0.709 ↓	0.738 [0.65, 0.83]	66, 77, 77	0.0026

3.1 The curve was noisy because the validation set was small ✗

The median epoch-to-epoch step of the validation AUROC is 0.0056 on the 75-gene draw, 0.0026 on the 158-gene split and 0.0020 on the 1,093-gene genome-wide split (Figure 2b). Two of the three width-24 cells of the previous protocol selected epochs 4 and 5 on validation peaks of 0.82 and 0.86 and scored 0.74 on the held-out genes; no cell on a split

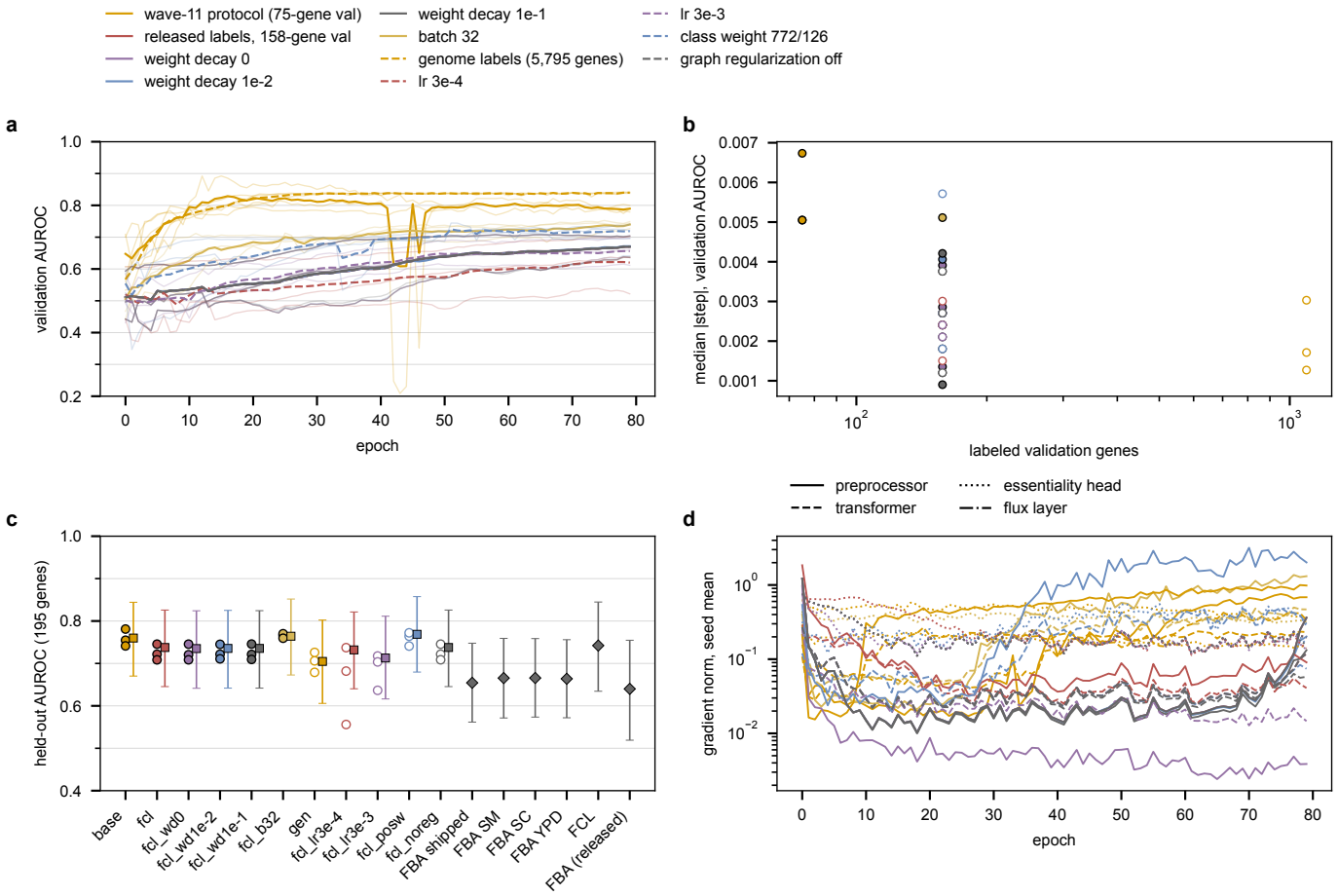


Figure 2: Round 1, the backbone. **a**, Validation AUROC per epoch, thin lines per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve against the number of labeled validation genes it was read on. **c**, Held-out AUROC on the 195 genes per cell, the seed mean, and the gene bootstrap of the pooled per-gene logits, beside FBA on each medium. **d**, Per-module gradient norm per epoch, seed mean, on the arms that logged it. Generated by `scripts/wave12_readout.py -round w12`.

file selects before epoch 50. The old protocol's curve reads near 0.8 on its own 75 genes and the split cells read 0.65 to 0.72 on theirs, so the levels are not comparable, only the steps are.

3.2 The split file costs training genes, and the cells are under-trained ✗

The split-file cell scores 0.725 against the old protocol's 0.759 on the same held-out genes. It trains on 629 labeled genes where the old draw trained on roughly 740, and every one of its seeds selected epoch 66 to 77 of 80 with the validation curve still rising (Figure 2a). At batch 128 an epoch is five optimizer steps, so 80 epochs is 400 updates. The two levers that move all three seeds in the same direction are both more optimization: batch 32, four times the steps per epoch, lifts the seeds by 0.020, 0.048 and 0.051 and collapses their spread to 0.005; the class weight, which steepens the curve, lifts them by 0.019, 0.051 and 0.033. Round 3 combines the two and raises the epoch budget.

3.3 Weight decay and graph regularization do nothing here ✗

Weight decay at 0, 10^{-2} and 10^{-1} reproduces the base cell to the third decimal on every seed, and so does removing graph regularization. At 400 updates with learning rate 10^{-3}, a decay of 10^{-1} shrinks a weight by four percent, so it is not a lever at this budget. Graph regularization is a different case: its configuration pins the nine gene graphs to attention heads 0 to 8 of the *second* transformer block, so on a one-block cell the term is never evaluated, and the arm with it off reproduces the arm with it on because both are the same cell. The term was inert on every one-block cell in this document; a two-block cell with four heads indexes a head it does not have and fails (round 8, job 2261), so the two-block cells run with it off. The learning rate at 3×10^{-4} and 3×10^{-3} is worse than 10^{-3} on the seed mean, with seed spreads of 0.09 and 0.04 that put the pooled numbers inside the base interval.

3.4 More labels from the genome make the held-out ranking worse ✗

The genome-wide file adds SGD's essential list and the viable single-mutant fitness records as nonessential, 4,368 training genes against 629, and scores 0.704 against 0.725, three of three seeds down. Its validation AUROC on its own 1,093 genes is 0.84 while the released held-out genes read 0.70, the widest validation-to-held-out gap of any arm. The viable-fitness negatives are a different population from the released split's nonessential genes, and a model that ranks them well does not rank the released genes better. Hypothesis, untested: the released split's nonessential class is enriched for genes with a measurable fitness defect, which the genome-wide negatives are not.

3.5 Gradient norms ✗

On the split-file cells the preprocessor's gradient norm starts near 10^{-2} and rises about thirty-fold over 80 epochs while the head's stays flat near 0.4 (Figure 2d); the clip at 10 never binds. The rise tracks the validation curve's climb, which is the same under-training reading from the gradient side.

3.6 Round 3: the two levers combined, a longer budget, and the fold ensemble ✗

Round 3 (jobs 2209 and 2224 to 2236) combines the two round-1 levers that moved three of three seeds, batch 32 and the class weight, at 80 and at 200 epochs, runs batch 32 alone at 200 epochs as the class weight's control, and trains five fold models on the cross-validation files, which partition the 787 labeled non-test genes five ways with the held-out genes fixed; their logits are averaged on the held-out genes, the protocol behind the published Flux Cone Learning number (Table 7, Figure 3). With the loader workers removed a 200-epoch cell takes 11 minutes.

Table 7: Round 3, more optimization and the fold ensemble. Batch 32 with the class weight at 80 and 200 epochs, batch 32 alone at 200 epochs, and the five fold models of the cross-validation files, each on its own fold. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 80 epochs	0.750	0.043	0.780, 0.769, 0.700	0.774 [0.69, 0.86]	75, 68, 31	0.0026
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
batch 32, 200 epochs	0.761	0.005	0.766, 0.761, 0.755	0.764 [0.67, 0.85]	74, 100, 181	0.0026
5-fold, fold 0	0.780	0.000	0.780	0.780 [0.69, 0.86]	75	0.0033
5-fold, fold 1	**0.806**	0.000	0.806	**0.806** [0.72, 0.89]	50	0.0036
5-fold, fold 2	0.797	0.000	0.797	0.797 [0.72, 0.87]	14	0.0033
5-fold, fold 3	0.751	0.000	0.751	0.751 [0.66, 0.84]	184	0.0021
5-fold, fold 4	0.789	0.000	0.789	0.789 [0.70, 0.87]	108	0.0039

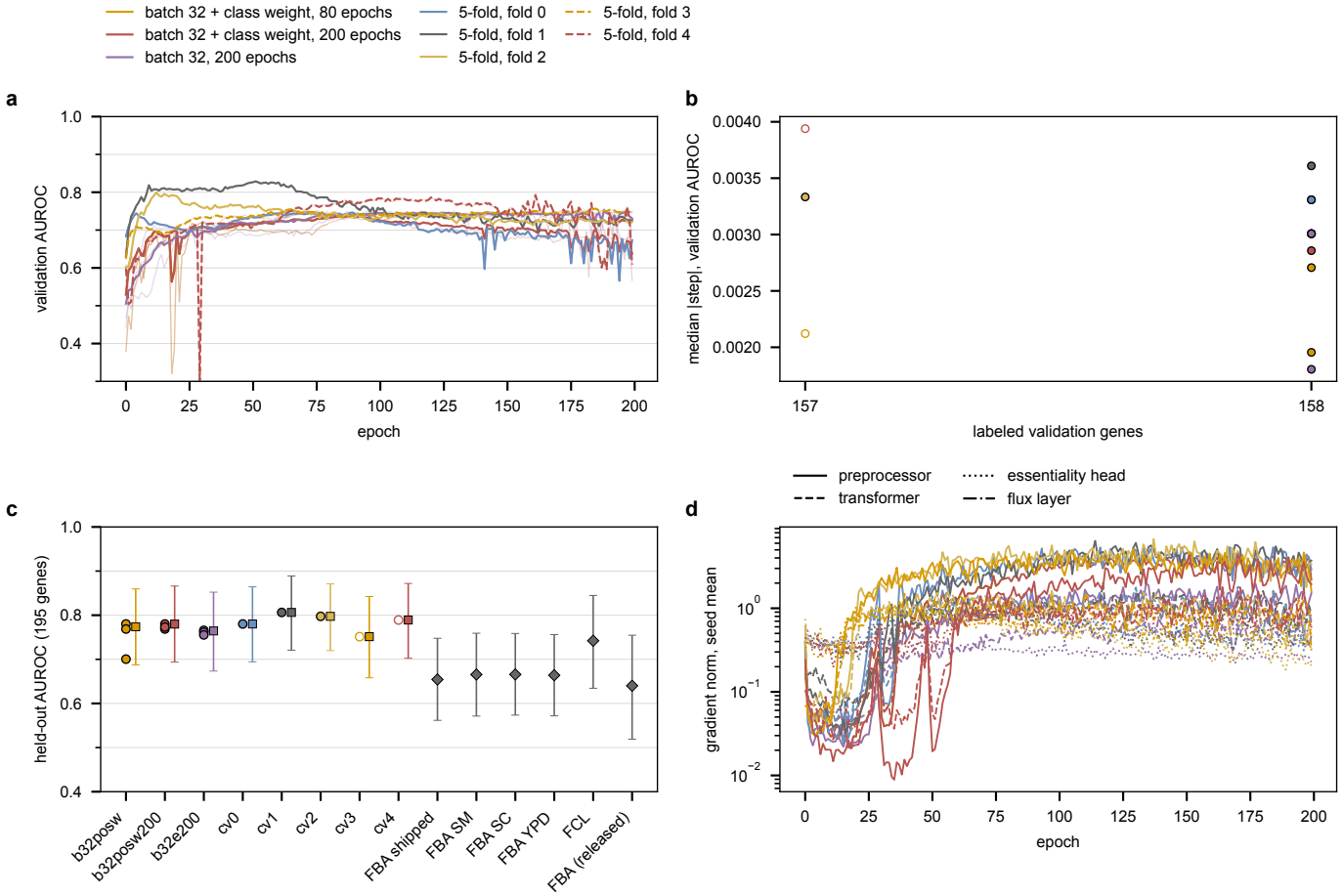


Figure 3: Round 3, the levers combined and the fold models. **a**, Validation AUROC per epoch, seed mean per arm; each fold model reads its own validation fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w14`.

Batch 32 with the class weight at 200 epochs scores 0.780, 0.769 and 0.773, a seed mean of 0.774 with a spread of 0.006, the tightest cell of any round; the pooled bootstrap is 0.780 [0.69, 0.87]. Batch 32 alone at 200 epochs scores 0.766, 0.761 and 0.755, so the class weight adds 0.01 to 0.02 on three of three seeds at the longer budget as it did at 80. The longer budget itself moves one seed: the 80- and 200-epoch cells share a trajectory through epoch 80 and select the same epoch on seeds 2700 and 2701 (75 and 68), while seed 2702 selects epoch 31 at 80 epochs and epoch 148 at 200, 0.700 against 0.773. Fold 0 of the cross-validation files is the released-label split file exactly (same 629 training, 158 validation and 195 held-out genes, both drawn at seed 0), so its fold model is the 200-epoch cell's first seed. The five fold models score 0.780, 0.806, 0.797, 0.751 and 0.789 on the held-out genes, a fold mean of 0.785, and their averaged logits score 0.795 [0.71, 0.88] (`cv_ensemble_readout.py`). That is the number to set beside the published 0.73 and beside 0.742 on these genes: the backbone under Flux Cone Learning's own protocol is 0.05 above it, and 0.01 to 0.04 below the ladder's rungs with FBA appended.

3.7 Round 5: FBA's growth ratios as an input ✗

The ladder's rungs cross 0.80 only with FBA's four growth ratios appended to the encoding, and no transformer cell has seen those numbers. Round 5 (jobs 2249 and 2250) gives the round-3 cell the same complement: the `fba` readout adds an affine of the deleted gene's knockout-to-wild-type growth ratio on the four media to the context head's logit, five parameters on top of the 16,561, with the four weights written into every epoch row. It runs at the three seeds, paired against the round-3 cell, and on the five cross-validation folds for the ensemble (Table 8, Figure 4). The read was fixed first: the transformer gains the ladder's complement if its seed mean moves from 0.774 toward 0.82 on three of three seeds and the fold ensemble from 0.795 toward the ladder's 0.81 to 0.83.

Table 8: Round 5, FBA's growth ratios as an input. The round-3 pooled cell and the same cell with an affine of FBA's four per-medium growth ratios for the deleted gene added to the context head, at three seeds, and that readout on the five cross-validation folds whose logits are averaged on the held-out genes. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844 ↑	0.838 [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratios, fold 0	0.814	0.000	0.814	0.814 [0.73, 0.89]	28	0.0033
context head + FBA ratios, fold 1	0.824	0.000	0.824	0.824 [0.74, 0.90]	42	0.0042
context head + FBA ratios, fold 2	0.839	0.000	0.839	0.839 [0.77, 0.91]	37	0.0048
context head + FBA ratios, fold 3	0.775	0.000	0.775	0.775 [0.69, 0.86]	122	0.0042
context head + FBA ratios, fold 4	**0.853**	0.000	0.853	**0.853** [0.78, 0.92]	54	0.0036

Both halves of the read hold. The cell scores 0.814, 0.805 and 0.844 against 0.780, 0.769 and 0.773 on the same seeds, up by 0.034, 0.037 and 0.071, a seed mean of 0.821 and a pooled bootstrap of 0.838 [0.76, 0.91]. The five fold models score 0.814, 0.824, 0.839, 0.775 and 0.853, a fold mean of 0.821, and their averaged logits 0.822 [0.74, 0.90] on the held-out genes, against 0.795 without the ratios. The transformer with the solver's four numbers sits where the ladder's best rungs sit (0.806 to 0.830), 0.08 above Flux Cone Learning on the same genes and 0.09 above its published number. The four weights end negative on every cell, largest on the shipped-bound and YPD columns (about −0.5 to −0.7 at the selected epoch against −0.3 to −0.5 on SM and near zero to −0.4 on SC), which is the direction a growth ratio must carry (a deletion that stops growth is essential), and the cells select earlier, at epochs 28 to 70 against 68 to 148: the affine is learned in the first epochs and the context head fills in around it.

The two sources are complementary and the complement is the same for the transformer as for a logistic regression on a frozen embedding: the backbone alone reads 0.77, the solver alone 0.66, and the sum 0.82. What the flux layer could not supply in rounds 2 and 4, where its growth column matched the head and its k_{cat} content went unread, the FBA solve supplies as four numbers, which locates the module's value on this endpoint in the solve itself rather than in the learned layer.

3.8 Round 8: width and depth on the FBA cell ✗

The round-5 cell ties the ladder's best rungs and does not pass them, and it is the smallest transformer in the document. Round 8 (jobs 2259, 2260, 2262 and 2263) moves its size one axis at a time with the readout, batch, class weight and budget fixed: width 24 and width 48 at one block (40,902 and 112,902 parameters against 16,566), two blocks at width 12 (18,450), and the two-layer preprocessor (542,480, most of it in the preprocessor's first layer), each at the three seeds, three cells packed on one card (Table 9, Figure 5). The read was fixed first: a cell passes the ladder if its seed mean clears 0.83 with three of three seeds above the round-5 cell.

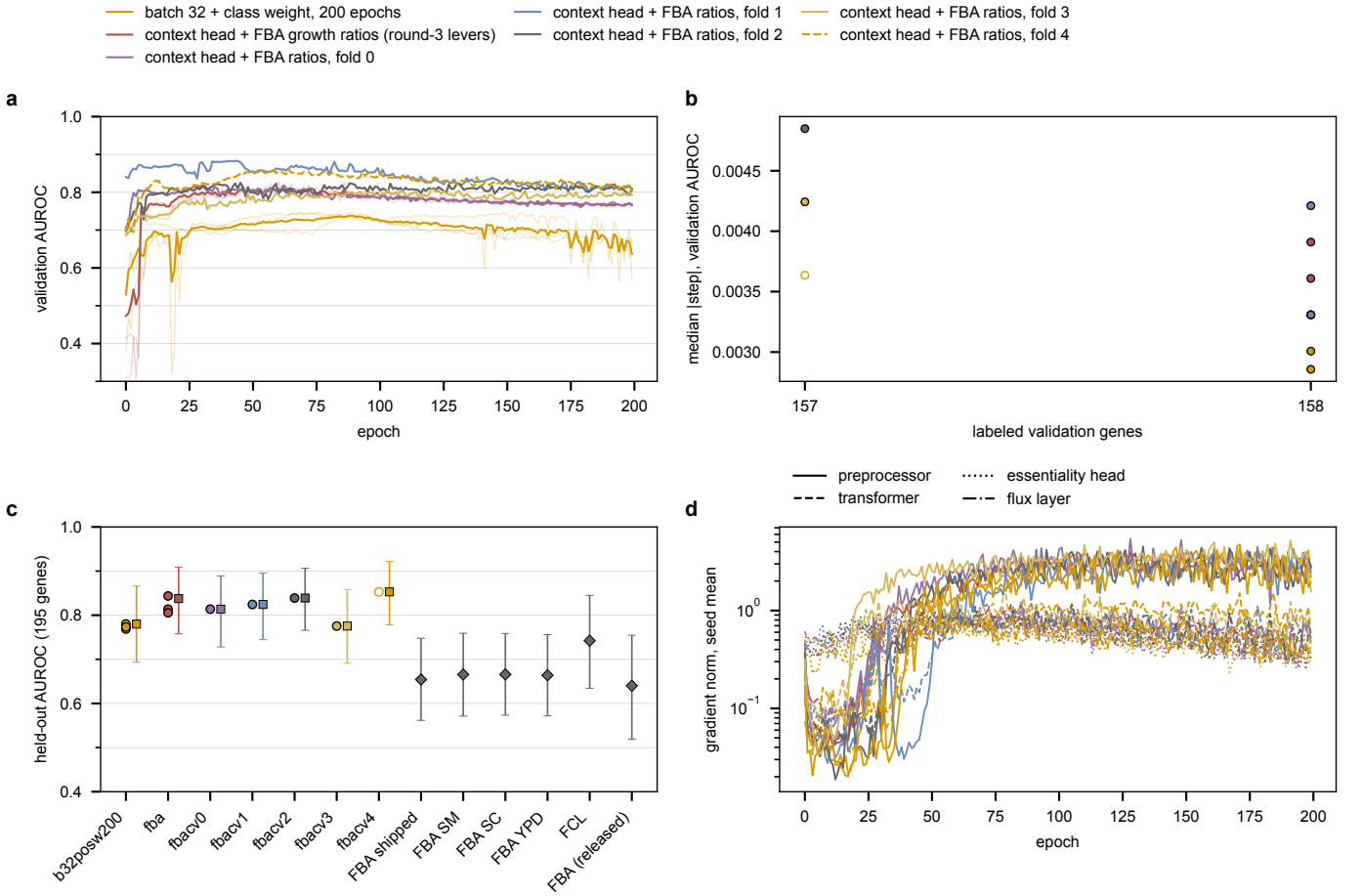


Figure 4: Round 5, FBA's growth ratios as an input. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w16`.

Table 9: Round 8, width and depth on the FBA cell. The round-5 cell (width 12, one block, one-layer preprocessor, context head plus the FBA affine) against the same readout at width 24 and 48, at two blocks, and with the two-layer preprocessor. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratios, width 24	0.817	0.005	0.818, 0.822, 0.812	0.823 [0.74, 0.90]	50, 41, 66	0.0045
context head + FBA ratios, width 48	**0.824**	0.004	0.827, 0.819, 0.826	0.831 [0.74, 0.91]	58, 86, 64	0.0037
context head + FBA ratios, two blocks	0.812	0.007	0.810, 0.820, 0.806	0.827 [0.74, 0.90]	57, 144, 41	0.0032
context head + FBA ratios, two-layer preprocessor	0.821	0.014	0.804, 0.829, 0.828	0.820 [0.74, 0.90]	82, 73, 103	0.0029

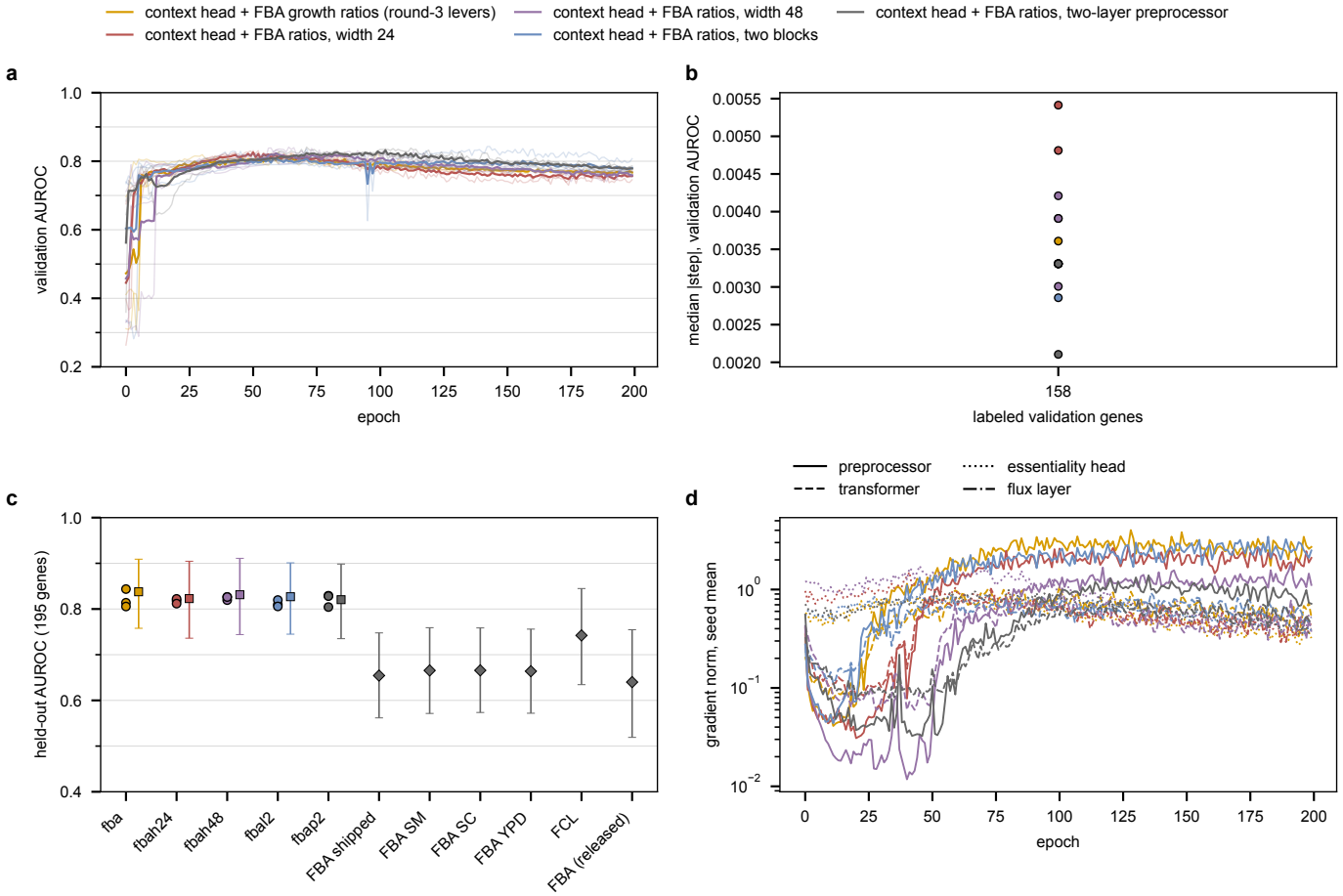


Figure 5: Round 8, width and depth on the FBA cell. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w19`.

No cell passes. Width 48 has the highest seed mean of the four, 0.824 against 0.821, with a seed spread of 0.004, and it is up on seeds 2700 and 2701 by 0.013 and 0.014 and down on seed 2702 by 0.018; width 24 scores 0.817, the two-layer preprocessor 0.821, and two blocks 0.812, the last down on two of three seeds. Seed 2702 of the round-5 cell, 0.844, is the highest single seed in the document and every round-8 cell is below it on that seed, so no arm clears the bar on all three. The seed spread narrows with width, from 0.020 at width 12 to 0.005 at 24 and 0.004 at 48, and every round-8 seed mean lies inside the round-5 cell's pooled bootstrap of 0.84 [0.76, 0.91]. The validation curves peak at 0.80 to 0.82 between epochs 40 and 90 on every arm and drift down from there, most at width 24, and the cells select epochs 41 to 144. The two-block cells run with graph regularization off, since the term indexes a second-block head that a four-head block does not have (the failed job 2261); it was inert on every one-block cell, so the comparison is like for like.

A width-12 transformer with FBA's four numbers reads 0.82 on these genes and neither four times the width, a second block, nor a wider preprocessor moves it by more than the seed noise. Capacity is not the lever on this endpoint and this split; the levers left untested are the gene encoding the transformer reads and what the solve supplies beyond four growth ratios.

3.9 Round 9: the FBA cell on the ladder's other encodings ✗

Every transformer cell so far reads ProtT5, and the ladder's best selected rung reads something else: logistic regression on the 197-dimensional chromosome-pathway indicator with the four ratios scores 0.830. Round 9 (jobs 2280 to 2283) opens the round-5 cell on the ladder's other encodings through `-node-embeddings`, with the readout, width, batch, class weight and budget fixed: the chromosome-pathway indicator alone, the indicator concatenated with ProtT5, ESM2, and the ladder's seven-encoding concatenation (ProtT5, CaLM, the two species-LM embeddings, the 5,979-base window, codon frequency and the indicator), each at the three seeds (Table 10, Figure 6; *chrom-pathway* in the figure is the chromosome-pathway indicator). The read is the round-8 one: a cell passes the ladder if its seed mean clears 0.83 with three of three seeds above the round-5 cell.

Table 10: Round 9, the FBA affine on the ladder's other encodings. The round-5 cell (ProtT5 backbone, context head plus the FBA affine) against the same cell opened on the chromosome-pathway indicator, on the indicator concatenated with ProtT5, on ESM2, and on the ladder's seven-encoding concatenation. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway backbone + FBA ratios	0.781	0.027	0.810, 0.759, 0.774 ↓	0.823 [0.75, 0.90]	21, 2, 2	0.0083
chrom-pathway + ProtT5 backbone + FBA ratios	**0.871**	0.015	0.860, 0.866, 0.888 ↑	**0.868** [0.80, 0.93]	53, 51, 38	0.0030
ESM2 backbone + FBA ratios	0.740	0.008	0.734, 0.736, 0.749 ↓	0.737 [0.64, 0.83]	133, 192, 64	0.0060
seven-encoding backbone + FBA ratios	0.820	0.009	0.822, 0.810, 0.827	0.823 [0.74, 0.90]	91, 92, 55	0.0026

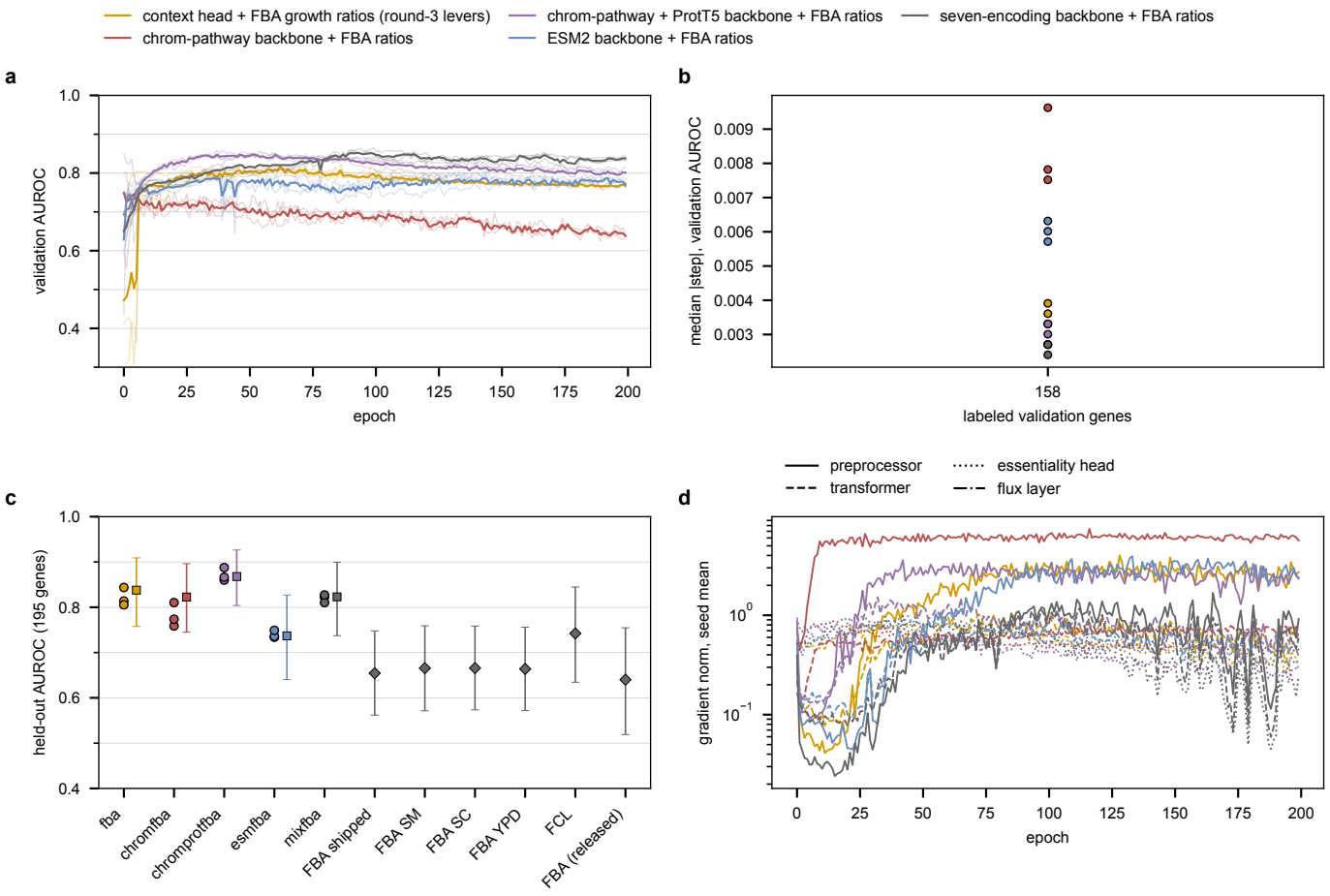


Figure 6: Round 9, the FBA cell on the ladder's other encodings. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w20`.

The indicator concatenated with ProtT5 passes. It scores 0.860, 0.866 and 0.888 against the round-5 cell's 0.814, 0.805 and 0.844 on the same seeds, up by 0.046, 0.061 and 0.044, a seed mean of 0.871 with a spread of 0.015 and a pooled

bootstrap of 0.868 [0.80, 0.93]; the lower end of that interval sits at the ladder's best rung. It is the first cell in this document whose seed mean clears 0.83 and whose weakest seed clears the round-5 cell's strongest. Its cells select epochs 38 to 53 on a validation curve that peaks near 0.85 between epochs 25 and 50 and drifts to 0.80 by epoch 200 (Figure 6a), and the cell has 18,930 parameters, 2,364 more than the round-5 cell for the wider input. The indicator alone does not: 0.810, 0.759 and 0.774, a mean of 0.781, with two seeds selecting epoch 2 and a validation curve that falls from 0.73 from the first epochs while its preprocessor's gradient norm sits near 5 for the whole run (Figure 6d), against 0.1 to 3 on the other arms. The two encodings add through the transformer, 0.87 against 0.83 for the indicator's rung and 0.82 for ProtT5's; the ladder never fit the pair on its own. Beside the chromosome and pathway indicators the encoding holds five continuous gene properties, two of them the gene's expression level and dispersion, and round 14 below says which columns carry the gain. ESM2 with the affine scores 0.740, below ProtT5 with the affine by 0.08 and level with ESM2's own ladder rung, and the seven-encoding concatenation 0.820, the round-5 cell's number from 78,066 parameters: adding five encodings to the two that pass returns the concatenation to where ProtT5 alone sits. Round 12 confirms the passing cell under the fold protocol and on four more seeds before the number is reported, since 0.87 on 195 genes is one cell's draw.

3.10 Round 12: the passing cell confirmed, and the number to report ✗

Round 12 (jobs 2291 to 2293) runs the round-9 cell at seeds 2703 to 2706 on the split file and as five fold models on the cross-validation files, the protocol behind the published Flux Cone Learning number, beside the round-5 cell's own fold models (Table 11, Figure 7). The four new seeds score 0.846, 0.877, 0.868 and 0.878, so over seven seeds the cell reads 0.869 with a spread of 0.014 and a pooled bootstrap of 0.870 [0.81, 0.93]; the weakest of the seven is above the round-5 cell's strongest. The five fold models score 0.860, 0.899, 0.889, 0.876 and 0.884 on the held-out genes, a fold mean of 0.881, every fold above the best of the round-5 folds (0.853), and their averaged logits score 0.893 [0.84, 0.94] (`cv_ensemble_readout.py -prefix w23_chromprotcv`), against 0.822 for the round-5 folds and 0.795 for the backbone alone. The fold models select epochs 25 to 53.

Table 11: Round 12, the passing cell confirmed. The round-5 cell (ProtT5 backbone, context head plus the FBA affine) and its five fold models, against the round-9 cell (chromosome-pathway indicator concatenated with ProtT5, same readout) on seven seeds, 2700 to 2706, and on the five cross-validation folds whose logits are averaged on the held-out genes. Width 12, one block, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway + ProtT5 backbone + FBA ratios	0.869	0.014	0.846 to 0.888 (7)	0.870 [0.81, 0.93]	38 to 53	0.0027
context head + FBA ratios, fold 0	0.814	0.000	0.814	0.814 [0.73, 0.89]	28	0.0033
context head + FBA ratios, fold 1	0.824	0.000	0.824	0.824 [0.74, 0.90]	42	0.0042
context head + FBA ratios, fold 2	0.839	0.000	0.839	0.839 [0.77, 0.91]	37	0.0048
context head + FBA ratios, fold 3	0.775	0.000	0.775	0.775 [0.69, 0.86]	122	0.0042
context head + FBA ratios, fold 4	0.853	0.000	0.853	0.853 [0.78, 0.92]	54	0.0036
chrom-pathway + ProtT5 + FBA ratios, fold 0	0.860	0.000	0.860	0.860 [0.79, 0.92]	53	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 1	**0.899**	0.000	0.899	**0.899** [0.84, 0.95]	32	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 2	0.889	0.000	0.889	0.889 [0.83, 0.94]	25	0.0055
chrom-pathway + ProtT5 + FBA ratios, fold 3	0.875	0.000	0.875	0.875 [0.81, 0.93]	34	0.0045
chrom-pathway + ProtT5 + FBA ratios, fold 4	0.884	0.000	0.884	0.884 [0.82, 0.94]	27	0.0055

The number to set beside the published 0.73 and beside 0.742 on these genes is the fold ensemble, 0.893 [0.84, 0.94]: a width-12 transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, scored under Flux Cone Learning's own protocol on its own held-out genes, is 0.15 above it, and its interval excludes the ladder's best rung (0.830). The cell has 18,930 parameters. Every part of the gain is accounted for by an input the ontology computes for every gene at no training cost: the four solver numbers (0.774 to 0.821 on the seed mean) and the pathway indicator concatenated with the protein embedding (0.821 to 0.869).

3.11 Round 11: the affine widened with the solve ✗

The ladder's solve-derived encodings (Section 2) carry 0.15 AUROC beyond the growth ratio. Round 11 (jobs 2294 to 2296) appends them to the round-5 cell's affine, standardized column by column over the labeled genes: the eight

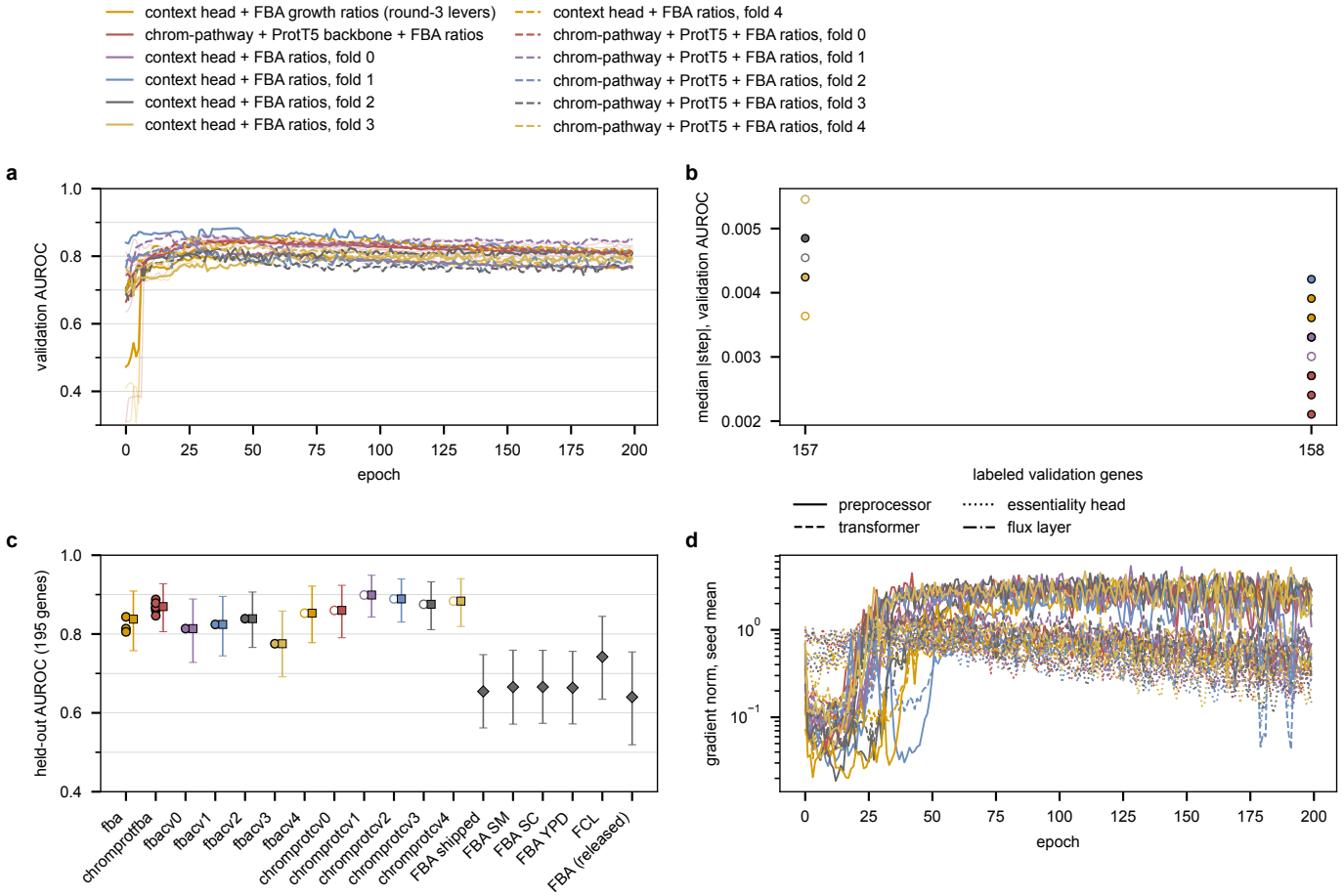


Figure 7: Round 12, the passing cell confirmed. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w23`.

reaction statistics (12 columns with the ratios), the deletion's pFBA flux vector (4,135), and both (4,139), each at the three seeds on the ProtT5 backbone (Table 12, Figure 8), under the round-8 read.

Table 12: Round 11, the affine widened with the solve. The round-5 cell (context head plus an affine of the four FBA growth ratios) against the same cell whose affine also reads the eight statistics of the deleted gene's reactions under the wild-type pFBA solve and its variability, the deletion's own pFBA flux vector (4,131 columns), or both; every column standardized over the labeled genes. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratios + eight reaction statistics	0.795	0.019	0.816, 0.790, 0.779	0.809 [0.70, 0.90]	55, 71, 124	0.0049
context head + FBA ratios + deletion pFBA vector	0.831	0.076	0.876, 0.743, 0.874	0.853 [0.76, 0.93]	70, 7, 99	0.0064
context head + FBA ratios + statistics + pFBA vector	**0.858**	0.008	0.850, 0.858, 0.867 ↑	**0.858** [0.77, 0.93]	188, 153, 162	0.0056

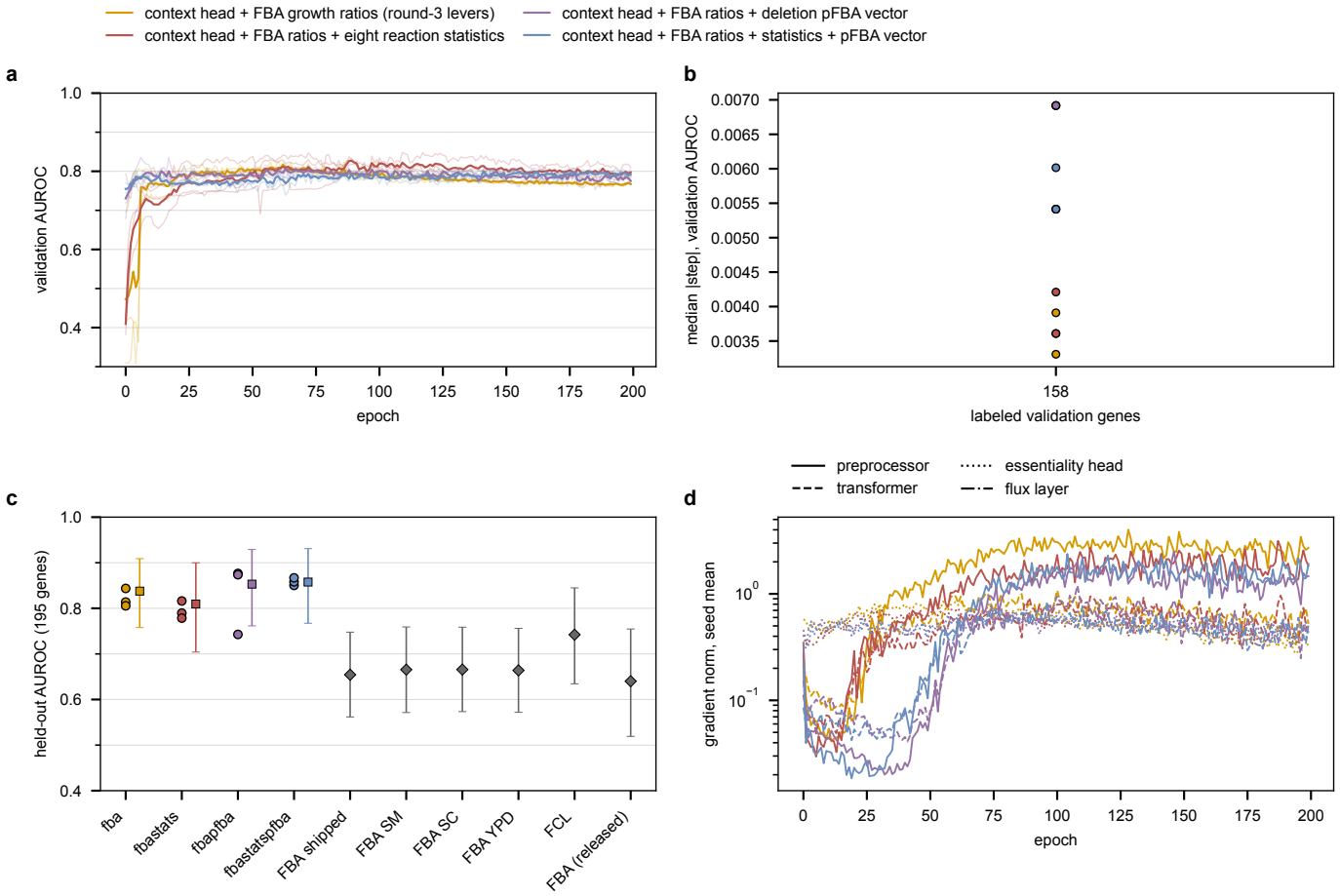


Figure 8: Round 11, the affine widened with the solve. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w22`.

The statistics and the vector together pass; neither does alone. With both, the cell scores 0.850, 0.858 and 0.867 against 0.814, 0.805 and 0.844, up on three of three seeds by 0.036, 0.053 and 0.023, a seed mean of 0.858 with a spread of 0.008, and it selects late, at epochs 153 to 188, on a validation curve that is still level at 200. The eight statistics alone read 0.816, 0.790 and 0.779, down on three of three seeds, though a random forest on the same twelve numbers scores 0.817 on the ladder: through a linear affine the statistics do not add to what the context head already carries, and the random forest's gain on them is the tree's. The vector alone reads 0.877, 0.743 and 0.874, two seeds at the level of the pair and one collapsed at epoch 7, so 4,131 columns on an affine over 629 training genes are unstable without the eight statistics beside them. Round 16 puts the passing affine on the round-9 backbone.

3.12 Round 14: which columns of the indicator carry the gain ✗

The chromosome-pathway indicator is three things: five scaled gene properties, a chromosome one-hot, and a multi-hot over 173 SGD pathways. Round 14 (jobs 2307 and 2308) drops the pathway columns (`normalized_chrom`, 24 dimensions) and runs the round-9 cell with and without ProtT5 beside it (Table 13, Figure 9).

Table 13: Round 14, attribution inside the chromosome-pathway indicator. The round-5 cell, the round-9 cells on the indicator alone and with ProtT5, and the same two cells on the indicator without its pathway columns (`normalized_chrom`: the five gene properties and the chromosome one-hot), all with the FBA affine. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway backbone + FBA ratios	0.781	0.027	0.810, 0.759, 0.774 ↓	0.823 [0.75, 0.90]	21, 2, 2	0.0083
chrom-pathway + ProtT5 backbone + FBA ratios	**0.871**	0.015	0.860, 0.866, 0.888 ↑	**0.868** [0.80, 0.93]	53, 51, 38	0.0030
gene properties + chromosome backbone + FBA ratios	0.669	0.030	0.636, 0.676, 0.694 ↓	0.663 [0.54, 0.77]	32, 69, 12	0.0110
gene properties + chromosome + ProtT5 backbone + FBA ratios	0.812	0.003	0.811, 0.816, 0.810	0.812 [0.72, 0.89]	65, 55, 53	0.0039

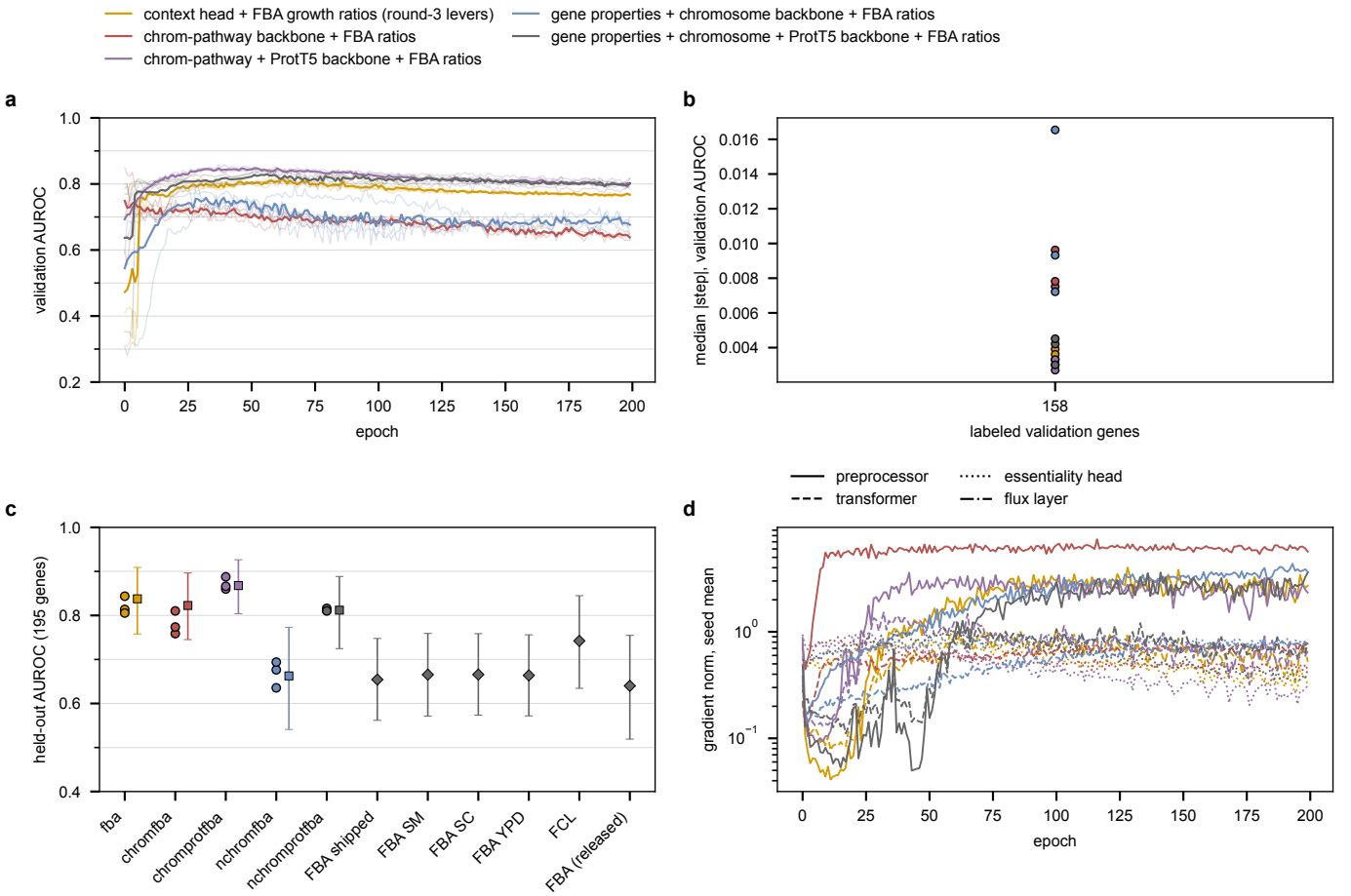


Figure 9: Round 14, attribution inside the chromosome-pathway indicator. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w25`.

The pathways carry it. Without them, the properties and the chromosome beside ProtT5 read 0.811, 0.816 and 0.810, a seed mean of 0.813 against 0.871 with the pathways and 0.821 for ProtT5 alone, so the five properties and the chromosome add nothing to the protein embedding, expression included; on their own they read 0.669, the solver's level. The 173 pathway columns are what lifts the cell from 0.82 to 0.87, and they do it only with ProtT5 beside them: the full indicator alone reads 0.78 (round 9). A gene's pathway membership and its protein sequence are complementary inputs on this endpoint, and each is cheap.

3.13 Round 16: the round-9 backbone with the round-11 affine ✗

Round 16 (jobs 2309 and 2310) puts the widened affine on the round-9 backbone: the chromosome-pathway indicator with ProtT5, and the four ratios with the eight statistics and the deletion flux vector on the logit, and the same without the statistics, three seeds each, against round 9 (Table 14, Figure 10).

Table 14: Round 16, the round-9 backbone with the round-11 affine. The round-5 cell, the round-9 cell (chromosome-pathway indicator with ProtT5), the round-11 cell (ProtT5 with the widened affine), and the round-9 backbone with the widened affine: the four ratios and the pFBA vector, and those plus the eight statistics. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway + ProtT5 backbone + FBA ratios	0.871	0.015	0.860, 0.866, 0.888 ↑	0.868 [0.80, 0.93]	53, 51, 38	0.0030
context head + FBA ratios + statistics + pFBA vector	0.858	0.008	0.850, 0.858, 0.867 ↑	0.858 [0.77, 0.93]	188, 153, 162	0.0056
chrom-pathway + ProtT5 backbone + FBA ratios + pFBA vector	0.828	0.068	0.769, 0.812, 0.902	0.867 [0.79, 0.93]	6, 13, 68	0.0027
chrom-pathway + ProtT5 backbone + ratios + statistics + pFBA	**0.880**	0.018	0.863, 0.878, 0.898 ↑	**0.889** [0.83, 0.94]	60, 38, 42	0.0029

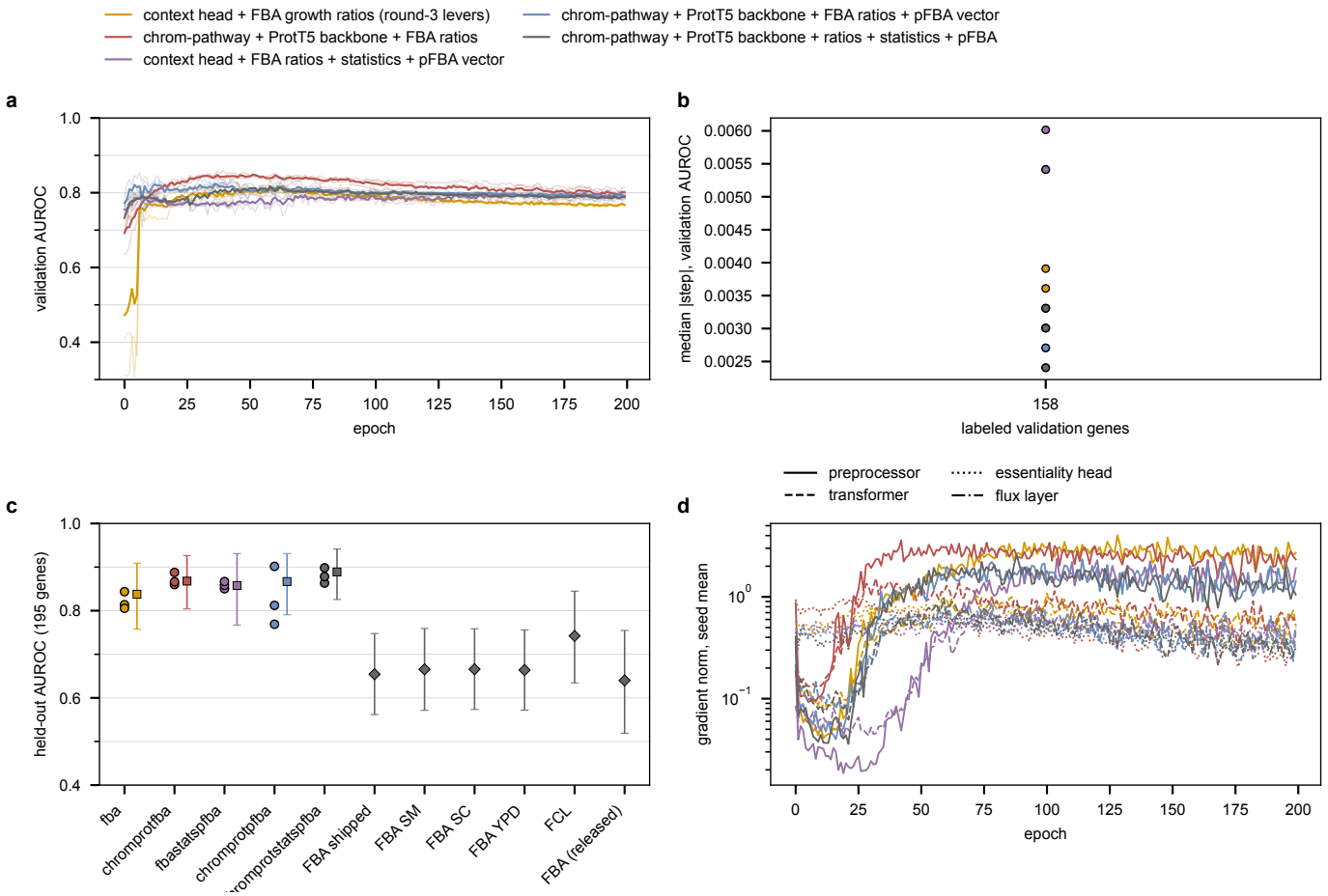


Figure 10: Round 16, the round-9 backbone with the round-11 affine. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w27`.

The two gains add, by a little. With the statistics and the vector the cell reads 0.863, 0.878 and 0.898 against round 9's 0.860, 0.866 and 0.888, up by 0.003, 0.012 and 0.010 on three of three seeds, a seed mean of 0.880 with a spread of 0.018 and a pooled bootstrap of 0.889 [0.83, 0.94]. The step is a fifth of the one the indicator gave and inside the seed spread, so it is the direction rather than the size that holds. Without the statistics the vector alone reads 0.769, 0.812 and 0.902, two seeds selecting epochs 6 and 13, the instability round 11 showed on the ProtT5 backbone, and the same cure. Round 17 runs the passing cell on the five folds, with the read fixed first: its ensemble replaces the round-12

ensemble's 0.893 as the reported number only if its fold models are above round 12's on at least four of the five folds; otherwise 0.893 stands.

3.14 Round 17: the round-16 cell on the five folds ✗

Round 17 (jobs 2312 and 2313) trains the round-16 cell on the five cross-validation files (Table 15, Figure 11). Its fold models score 0.863, 0.860, 0.887, 0.841 and 0.882 on the held-out genes, a fold mean of 0.867, against round 12's 0.860, 0.899, 0.889, 0.876 and 0.884: above on fold 0 by 0.003 and below on the other four, by 0.002 to 0.039. Their averaged logits score 0.901 [0.85, 0.95], above the round-12 ensemble's 0.893 by 0.008, with the two intervals almost coincident.

Table 15: Round 17, the round-16 cell on the five folds. The round-9 and round-16 cells on the split file, the round-12 fold models of the round-9 cell, and the five fold models of the round-16 cell (chromosome-pathway indicator with ProtT5, the affine reading the four ratios, the eight statistics and the pFBA vector), whose logits are averaged on the held-out genes. Width 12, one block, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
chrom-pathway + ProtT5 backbone + FBA ratios	0.871	0.015	0.860, 0.866, 0.888	0.868 [0.80, 0.93]	53, 51, 38	0.0030
chrom-pathway + ProtT5 backbone + ratios + statistics + pFBA	0.880	0.018	0.863, 0.878, 0.898 ↑	0.889 [0.83, 0.94]	60, 38, 42	0.0029
chrom-pathway + ProtT5 + FBA ratios, fold 0	0.860	0.000	0.860	0.860 [0.79, 0.92]	53	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 1	**0.899**	0.000	0.899	**0.899** [0.84, 0.95]	32	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 2	0.889	0.000	0.889	0.889 [0.83, 0.94]	25	0.0055
chrom-pathway + ProtT5 + FBA ratios, fold 3	0.875	0.000	0.875	0.875 [0.81, 0.93]	34	0.0045
chrom-pathway + ProtT5 + FBA ratios, fold 4	0.884	0.000	0.884	0.884 [0.82, 0.94]	27	0.0055
chrom-pathway + ProtT5 + widened affine, fold 0	0.863	0.000	0.863	0.863 [0.79, 0.93]	60	0.0024
chrom-pathway + ProtT5 + widened affine, fold 1	0.860	0.000	0.860	0.860 [0.77, 0.93]	135	0.0078
chrom-pathway + ProtT5 + widened affine, fold 2	0.887	0.000	0.887	0.887 [0.81, 0.95]	26	0.0085
chrom-pathway + ProtT5 + widened affine, fold 3	0.840	0.000	0.840	0.840 [0.75, 0.92]	38	0.0052
chrom-pathway + ProtT5 + widened affine, fold 4	0.882	0.000	0.882	0.882 [0.82, 0.94]	42	0.0082

By the rule fixed before the round, 0.893 stands: the widened affine's fold models are below round 12's on four of five folds, and an ensemble 0.008 higher from fold models 0.014 lower on average is the kind of draw the rule exists to refuse. The two cells are one cell to the resolution of 195 genes, and the reported number for this document is the round-12 ensemble, 0.893 [0.84, 0.94], from the chromosome-pathway indicator with ProtT5 and the four FBA growth ratios.

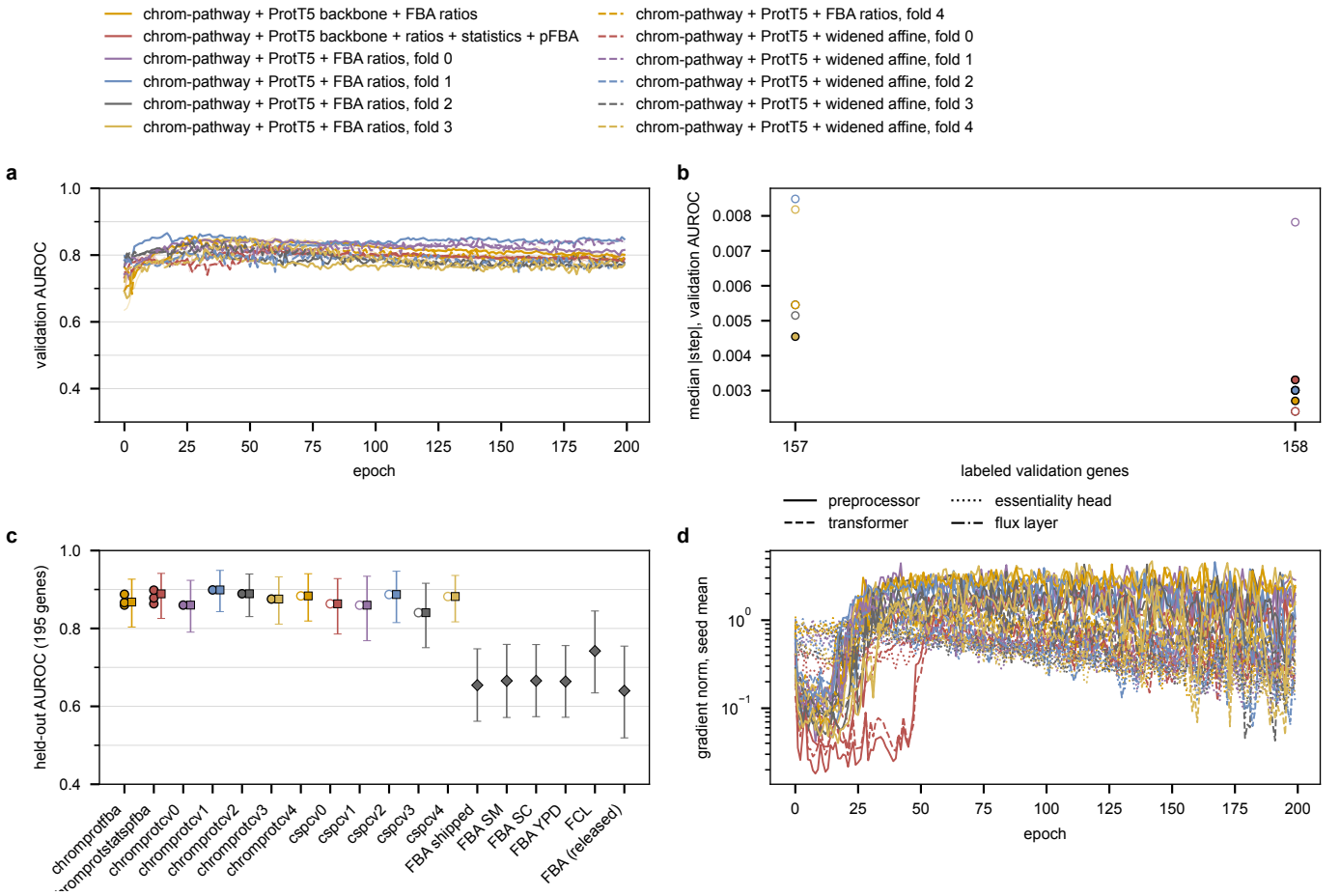


Figure 11: Round 17, the round-16 cell on the five folds. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w28`.

4 The metabolic module: the solve helps, so why not the layer? ✗

One linear program per deletion on yeast-GEM ranks essential genes at 0.65 on its own and lifts every model in this document by 0.05 when its four growth ratios are appended (Sections 2 and 3). The metabolic module is built on the same genome-scale model: the same stoichiometry, the same bounds, the same gene-protein-reaction rules closing the deleted gene's reactions. The question of this section is why routing the prediction through it does not do what the solve does. The rounds answer it in steps: the module neither helps nor hurts on the backbone (rounds 2 and 4); it adds nothing on top of the solve (round 6); its growth column is not growth but a free coordinate, because the layer never reaches the polytope the solve optimizes over (the probe, round 7); handed the solve's own point on that polytope, the anchored layer's constraint set discards it (round 10); and the gradient reaches the layer throughout, so none of this is a gradient-flow failure. The section closes with what the comparison says a layer would have to compute to help.

The pooled backbone reads its logit from an MLP on the perturbed context (Figure 13). A flux arm builds the enzyme-constrained thermodynamic flux layer on the same context and exposes one physically named quantity, the flux through the growth reaction `r_2111`. Three readouts put that quantity to different uses, and they differ in what the backbone's gradient has to pass through.

biomass (forced) $\ell = a\, v_{\text{growth}} + b$. The growth column is the only path from the context to the logit, so the module carries the whole prediction or the cell scores chance (Figure 14). This is FBA's own reading of essentiality with a learned threshold, and it is the wave-9 design.

context + growth $\ell = \text{MLP}(h) + a\, v_{\text{growth}} + b$. Both paths from the first step; the per-module gradient norms say how much of each update the module carries when it is not forced to carry all of it.

gated $\ell = \text{MLP}(h) + g\,(a\, v_{\text{growth}} + b)$ with $g_0 = 0$ (Figure 15). At initialization the cell is the pooled model exactly; the gate's trajectory, logged every epoch, says whether the optimizer chose the flux path, and the `flux_layer` gradient line says whether gradient reached the layer, which at $g = 0$ it cannot.



Figure 12: Color key for the readout diagrams. White is the input; orange a learned part of the backbone (the context, the abundance head); yellow the flux layer and its capacity terms, dashed where the box is a fixed table (the k_{cat} table); purple the fixed stoichiometry of the GEM; red the prediction and the loss. The key holds for Figures 13 to 15. Rendered from `notes/experiments.028-gene-essentiality.mermaid.readouts.md`.

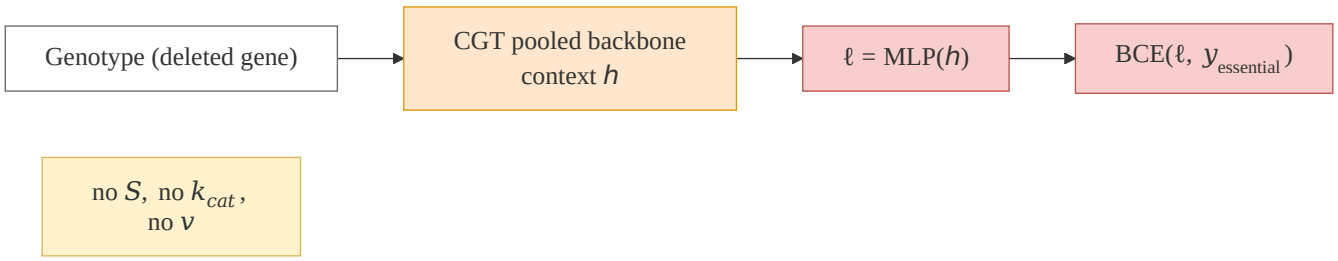


Figure 13: The pooled readout. The backbone alone: one MLP from the perturbed context to a logit, binary cross entropy against the released label. No stoichiometry, no kinetic parameters, no flux. Colors as in Figure 12.

4.1 What the layer's flux violates ✗

Every flux arm in this document builds the anchored layer: the predicted k_{cat} tier (97 measured, 782 homolog, 2,831 predicted, 18 default catalytic units), the union-external reaction energies (3,779 of 4,131 reactions with a $\Delta_r G'$), the enzyme-capacity ceiling from the abundance head in the box, and the second law, the protein budget and the Gibbs dissipation as penalty terms at the constraint scale. The layer logs every one of those residuals each epoch, and Table 16 reads them at the selected epoch across the module arms. With the balance exact, the flux is outside its bounds on 58 to 63 percent of reactions, runs against its reaction energy on 44 to 51 percent, uses 0.2 to 7 times the protein pool, and dissipates 5,000 to 12,000 J/gDW/h against the 3,700 limit, with the penalty on or off; with the bounds exact (round 7) the second-law violation falls to a third and the dissipation rises to 100,000. No arm's flux satisfies the constraints

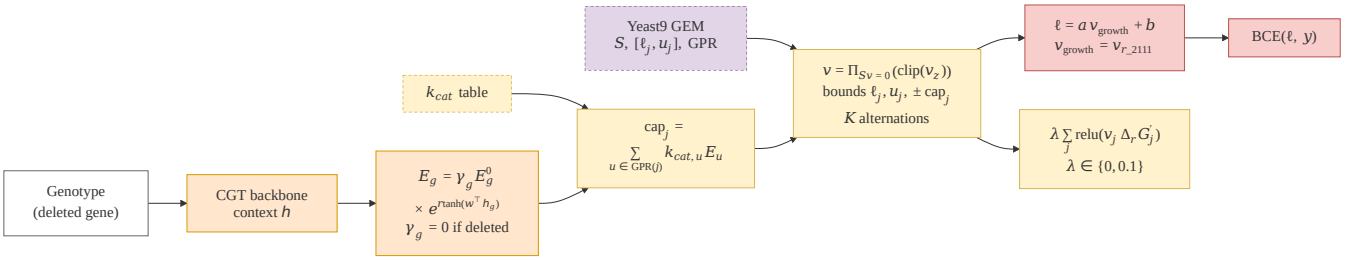


Figure 14: The forced biomass readout. The context sets each enzyme's abundance (a deleted gene's is zero), the k_{cat} table turns abundances into capacities, and the flux is the projection onto $Sv = 0$ of the capacity-clipped proposal; the logit is an affine of the growth column alone. The thermodynamic penalty is a loss on the flux with no path to the prediction, at scale 0.1 or 0. Colors as in Figure 12.

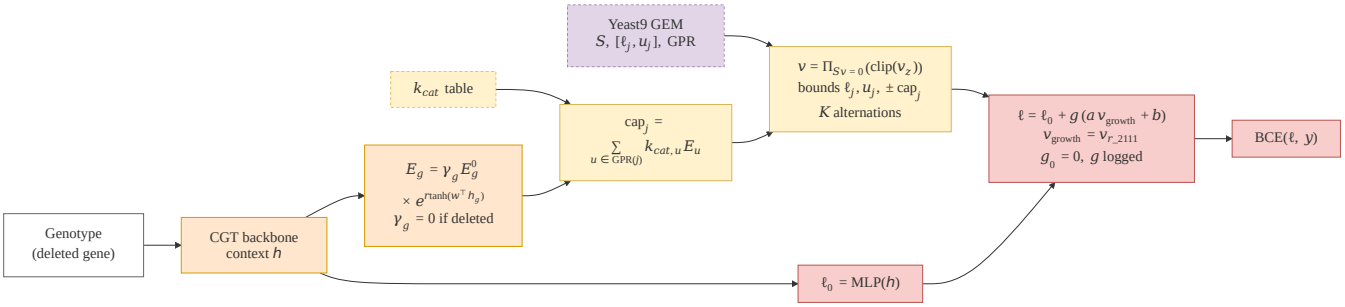


Figure 15: The gated readout. The pooled MLP's logit plus the growth affine behind a gate initialized at zero. The gate is a parameter of the readout; its value is written into every epoch row as `ess_gate`. Colors as in Figure 12.

the layer was built to hold, so on this endpoint the k_{cat} table and the reaction energies are inputs to a projection that does not honor them, which is the precise sense in which the module is not the solve.

Table 16: What the layer's flux violates at the selected epoch, seed mean per module arm: the median mass-balance residual $|Sv|$ over metabolites; the fraction of reactions outside their bounds; the fraction of reactions running against their reaction energy ($v_j\, \Delta_r G_j' > 0$); the enzyme budget used as a multiple of the protein pool; and the Gibbs dissipation in J/gDW/h against the 3,700 limit. Rounds are abbreviated r2 to r13. Every arm builds the anchored layer (predicted k_{cat} tier, union-external ΔG) unless it names the GEM box. Constraint scale 0 means no penalty on any of these terms; 0.1 penalizes all of them.

arm	cells	$\lvert Sv \rvert$ median	box	second law	budget	dissipation
r2 forced, scale 0.1	3	1.4e-08	0.63	0.48	0.2	-8,375
r2 forced, scale 0	3	1.3e-08	0.62	0.51	0.8	7,099
r2 gated, scale 0.1	3	1.1e-08	0.63	0.44	0.3	-9,854
r4 forced, scale 0	3	7.9e-09	0.59	0.51	2.4	10,264
r4 gated, scale 0	3	1.1e-08	0.58	0.51	3.6	12,277
r6 forced + FBA affine	3	7.5e-09	0.59	0.51	2.2	10,115
r7 forced, clip last	3	9.0e-01	0.00	0.32	1.3	102,065
r7 forced, box form	3	2.0e+00	0.00	0.38	1.4	120,815
r10 forced, warm start	3	1.3e-09	0.33	0.49	6.9	5,815
r10 gated, warm start	3	9.3e-10	0.36	0.49	1.3	5,104
r13 forced, GEM box, warm start	3	2.8e-09	0.26	0.00	nan	nan
r13 gated, GEM box, warm start	3	2.5e-09	0.27	0.00	nan	nan

4.2 Round 2: the module on the stable backbone, with and without the penalties ✗

Round 2 (jobs 2182 to 2208) puts the three readouts on the width-12 cell of Section 3, on the released-label split, at three seeds each: the forced and gated readouts at constraint scale 0.1 and 0, the context readout at both scales, the forced readout at scale 0 with the k_{cat} vector permuted across catalytic units (same architecture, no unit-to-value correspondence), and the stoichiometry-only layer (`flux_off`: no thermodynamics, no capacity) with the forced and context readouts. Every flux arm carries the wave-9 flux configuration otherwise: the predicted k_{cat} tier, the union-external thermodynamic source, twenty warmup epochs, the dataset medium, the mixture concentration space. The paired comparator is the round-1 pooled cell on the same split at the same batch, weight decay and budget (Table 17, Figure 16).

The reads were fixed before the numbers: *not hurt* is the gated arm against the pooled cell paired by seed, with the gate's trajectory beside it; *help* is the forced arm above the pooled cell on three of three seeds and above its shuffled-k_{cat} control; the *collapse's cause* is the forced arm at scale 0.1 against scale 0, since on the betaxanthin targets the penalty

scale was the dose that starved the layer's input.

Table 17: Round 2, the metabolic module. The round-1 pooled cell on the released-label split (`fcl`) and the same cell with the prediction routed through the flux layer: the growth column alone (forced), the context head plus the growth affine, and that affine behind a zero-initialized gate, at constraint scale 0.1 and 0, with the shuffled-kcat control and the stoichiometry-only layer. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708	0.738 [0.65, 0.83]	66, 77, 77	0.0026
growth column only, constraints 0.1	0.706	0.105	0.759, 0.585, 0.774	0.765 [0.67, 0.85]	77, 26, 76	0.0037
growth column only, constraints 0	0.732	0.068	0.770, 0.653, 0.772	**0.773** [0.68, 0.86]	72, 72, 72	0.0047
growth column only, kcat shuffled, constraints 0	**0.751**	0.029	0.771, 0.718, 0.765	0.769 [0.68, 0.86]	74, 77, 75	0.0029
context + growth column, constraints 0.1	0.666	0.086	0.566, 0.709, 0.722	0.717 [0.61, 0.82]	49, 77, 77	0.0053
context + growth column, constraints 0	0.667	0.096	0.556, 0.727, 0.718	0.724 [0.62, 0.82]	49, 73, 77	0.0068
gated growth column, constraints 0.1	0.719	0.029	0.721, 0.747, 0.690	0.738 [0.64, 0.83]	41, 77, 77	0.0027
gated growth column, constraints 0	0.735	0.018	0.714, 0.744, 0.747	0.747 [0.65, 0.84]	40, 77, 77	0.0031
growth column only, stoichiometry only	0.606	0.072	0.524, 0.631, 0.661 ↓	0.664 [0.56, 0.77]	19, 73, 65	0.0033
context + growth column, stoichiometry only	0.705	0.023	0.719, 0.717, 0.678 ↓	0.755 [0.67, 0.84]	75, 76, 61	0.0042

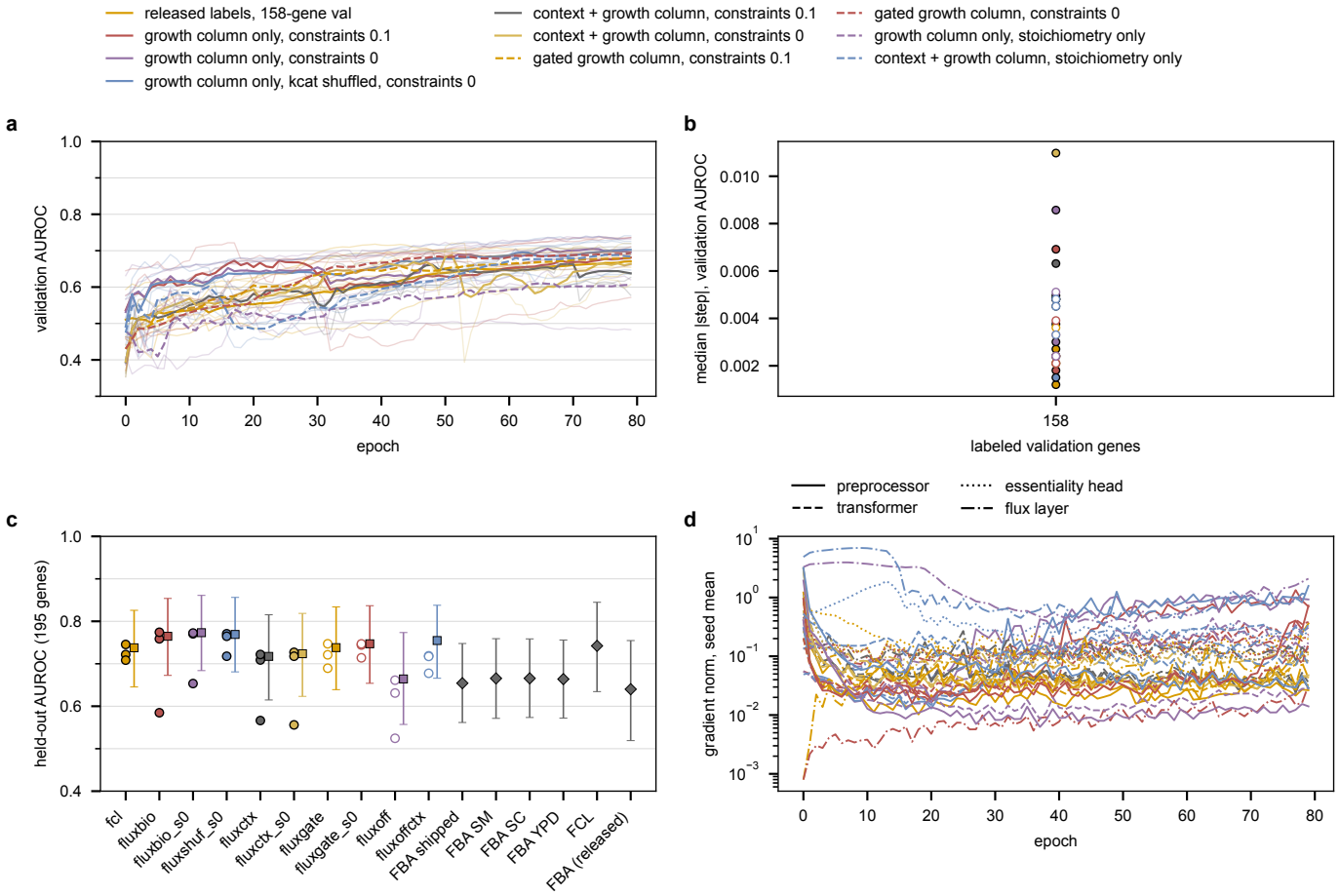


Figure 16: Round 2, the metabolic module. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve per cell, against the number of labeled validation genes, 158 for every cell in this round. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w13`.

Not hurt holds. The gated arm at scale 0 scores 0.714, 0.744 and 0.747 against the pooled cell's 0.745, 0.722 and 0.708 on the same seeds, two of three above by 0.02 to 0.04 and one below by 0.03, a seed mean of 0.735 against 0.725; at scale 0.1 the same comparison is one of three above. The gate leaves zero and stops: at epoch 80 it reads −0.056 to −0.063 on all six gated cells, and the flux layer's gradient norm sits at 0.01 to 0.07 against 0.22 to 0.36 on the forced arm's two seeds that train. The optimizer takes a small negative weight on the growth affine and the layer receives a

tenth of the forced arm's gradient, so the gated cell is a pooled cell with a small correction, and it scores like one.

Help does not hold. The forced arm at scale 0 scores 0.770, 0.653 and 0.772, two of three seeds above the pooled cell and one seed 0.07 below; at scale 0.1 the split is the same and the low seed reads 0.585. The shuffled-k_{cat} control, the same layer with the catalytic constants permuted across units, scores 0.771, 0.718 and 0.765, a seed mean of 0.751 against the real table's 0.732, so whatever the growth column carries here does not come from the correspondence between units and constants. The forced arm's pooled bootstrap, 0.773 [0.68, 0.86], is above Flux Cone Learning's 0.742 and FBA's 0.640 to 0.666 on the same genes, level with the pooled cell's 0.738, and below the ladder's 0.80 line.

The collapse on this endpoint is not the penalty's. Seed 2701 falls at both scales, to 0.585 at 0.1 and 0.653 at 0, and the two runs are the same run: the gradient norms start where the other seeds' do, decay within ten epochs to a hundredth of them (preprocessor 0.002, transformer 0.003 at epoch 80, against 1.0 and 0.5 on seed 2700) and stay there, with the training loss flat at 0.46 against 0.34, which is a stalled seed rather than a starved layer. The stoichiometry-only layer collapses on all three seeds (0.524, 0.631, 0.661), so the capacity bound and the thermodynamic term are what keep the growth column informative rather than what starve it, the opposite of the betaxanthin reading. Adding the context head beside the growth affine without a gate costs 0.06 on the seed mean (0.666 and 0.667 against 0.725) and puts one seed at 0.56; the same sum behind the zero-initialized gate loses nothing.

4.3 Round 4: the module on the best backbone ✗

Round 4 (jobs 2246 to 2248, three cells per GPU) repeats the gated readout, the forced readout and the shuffled-k_{cat} control at constraint scale 0 on the round-3 cell, batch 32 with the class weight at 200 epochs, paired by seed against it (Table 18, Figure 17). The reads are the round-2 reads.

Table 18: Round 4, the module on the round-3 levers. The round-3 pooled cell (batch 32, class weight, 200 epochs) and the same cell with the prediction routed through the flux layer at constraint scale 0: the gated readout, the forced growth column, and the forced growth column with the kcat vector shuffled. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
growth column only, kcat shuffled (round-3 levers)	**0.779**	0.006	0.785, 0.781, 0.772	**0.781** [0.69, 0.87]	66, 71, 74	0.0048

Every arm lands within 0.015 of the pooled cell on every seed. The gated readout scores 0.766, 0.782 and 0.765 against 0.780, 0.769 and 0.773, one of three above; the forced readout 0.772, 0.781 and 0.778, two of three above by 0.005 and 0.012 and one below by 0.008; the shuffled-k_{cat} control 0.785, 0.781 and 0.772, two of three above, a seed mean of 0.779 against the real table's 0.777. The gate now moves: it leaves zero after epoch 10 and climbs steadily to −0.17, +0.20 and +0.11 at epoch 200, of either sign, with the flux layer's gradient at 0.06 to 0.12 against 0.22 to 0.65 in the forced arm. The stalled seed of round 2 does not recur under batch 32 with the class weight. On every arm the validation curve peaks between epochs 60 and 110 and falls afterwards (Figure 17a), so the 200-epoch budget is more than the cell uses and the smoothed selection rule is what the number rests on.

The two rounds together answer the two questions. The module does not hurt: on the stable backbone and on the best one, the gated cell is the pooled cell to within seed noise. The module does not help: the forced growth column, which carries the whole prediction, matches the pooled MLP head to within 0.01, and it matches it equally well with the catalytic constants permuted, so the growth column is as expressive a readout as the head but its k_{cat} content is not what it reads. On this endpoint, at this width, the enzyme-constrained flux is a learned nonlinearity over the context rather than a physical prior that ranks genes.

4.4 Round 6: one medium against four, the layer on top of the solve, and two other embeddings ✗

Round 5 (Section 3) put FBA's four growth ratios beside the context head and reached the ladder's level. Round 6 (jobs 2251 to 2256, three cells per GPU) asks three questions of that result on the same cell, batch 32 with the class weight at 200 epochs, three seeds each, paired against the round-5 cell (Table 19, Figure 18): whether one medium's ratio carries what four do; whether the flux layer adds anything once the solve is present, with the forced growth column plus the affine (no context head) and the gated readout plus the affine, both at constraint scale 0; and whether the backbone on another node embedding does better, on codon frequency (64 dimensions, the encoding that reaches 0.82 with FBA

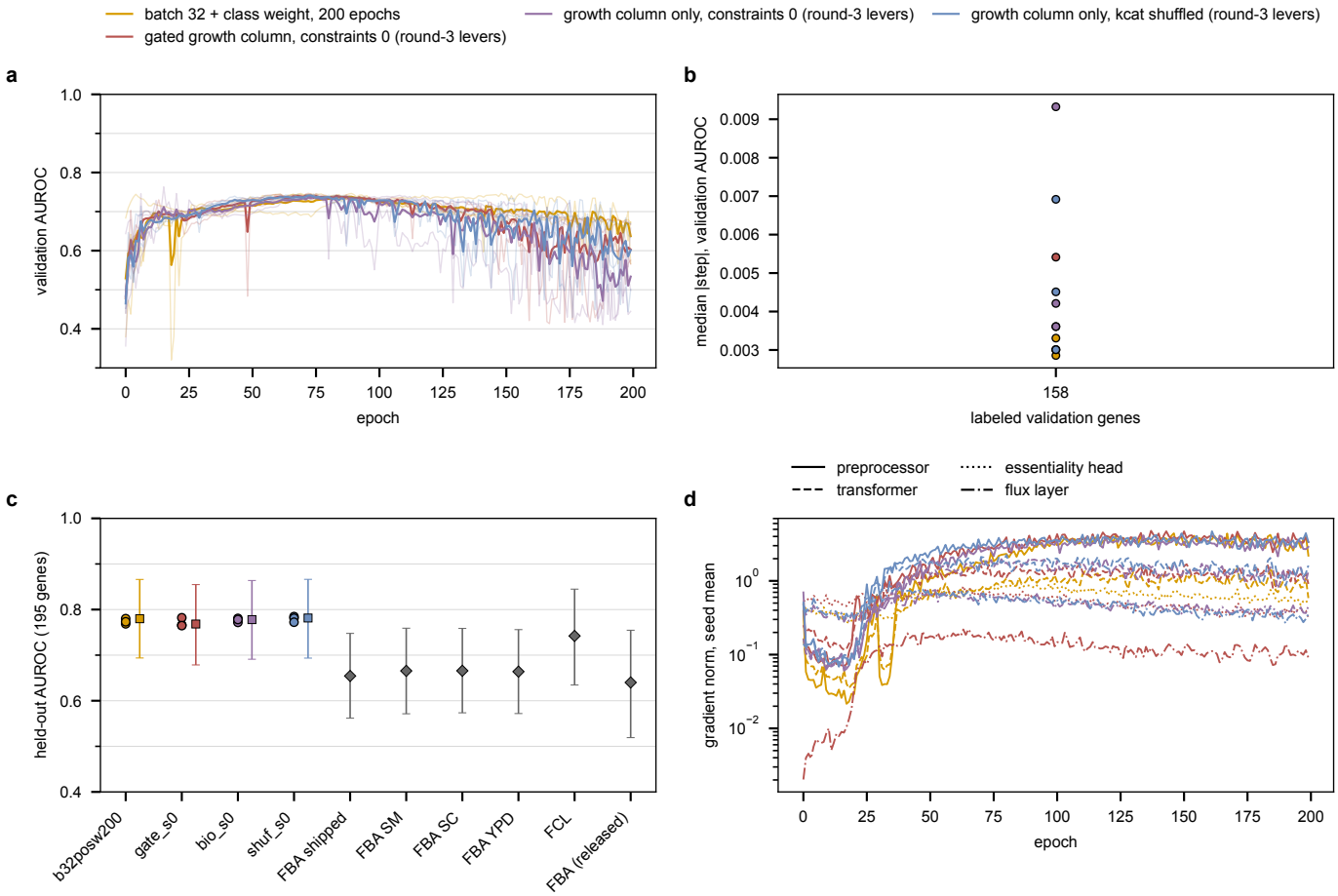


Figure 17: Round 4, the module on the round-3 cell. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 158 for every cell. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w15`.

on the ladder from a support vector classifier) and on the species-LM composite the expression line uses (the 5′ and 3′ species language model, ProtT5 and CaLM, 3,332 dimensions), with and without the affine.

Table 19: Round 6, one medium, the layer on top of the solve, and two other embeddings. The round-3 pooled cell and the round-5 FBA-readout cell, then: the FBA affine on the YPD ratio alone; the forced and the gated growth column of the flux layer with the FBA affine beside them (constraint scale 0); and the pooled backbone opened on codon frequency, and on the species-LM composite (5' and 3' species LM, ProtT5, CaLM), with and without the FBA affine. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratio, YPD only	0.795	0.022	0.819, 0.775, 0.789 ↑	0.808 [0.73, 0.89]	18, 98, 61	0.0034
growth column + FBA ratios, no context head	0.818	0.014	0.834, 0.811, 0.809 ↑	0.820 [0.74, 0.90]	74, 83, 57	0.0052
gated growth column + FBA ratios	**0.821**	0.011	0.810, 0.821, 0.832 ↑	0.827 [0.75, 0.90]	116, 43, 92	0.0041
codon-frequency backbone + FBA ratios	0.731	0.043	0.747, 0.682, 0.764 ↓	0.721 [0.62, 0.81]	16, 172, 14	0.0130
species-LM composite backbone + FBA ratios	0.816	0.013	0.802, 0.825, 0.822 ↑	0.819 [0.73, 0.90]	149, 65, 65	0.0031
species-LM composite backbone	0.778	0.005	0.783, 0.779, 0.773	0.776 [0.69, 0.85]	30, 46, 23	0.0035

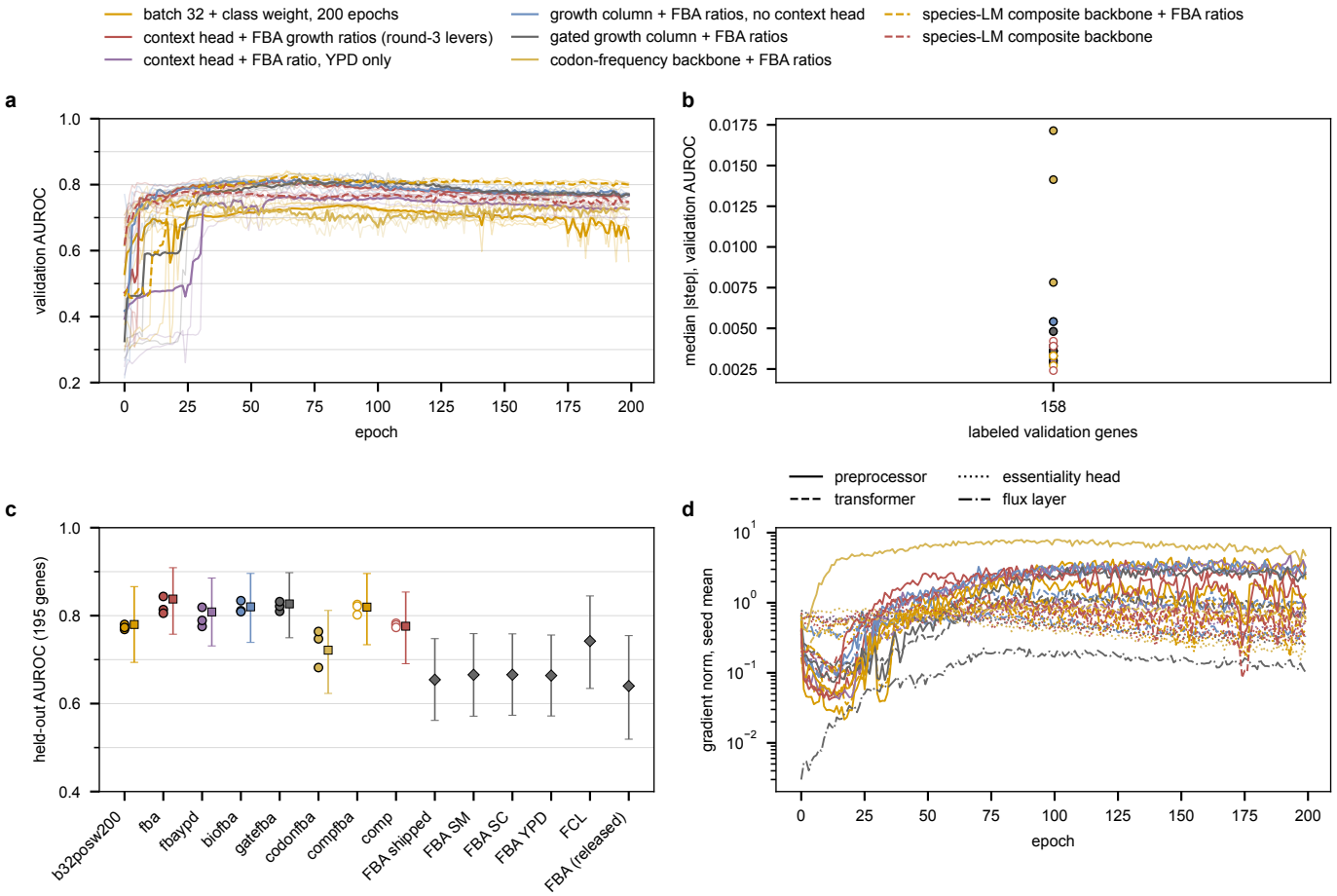


Figure 18: Round 6. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; the second tier of arms is dashed. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points, hollow for the second tier), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w17`.

Four media carry more than one. The YPD ratio alone scores 0.819, 0.775 and 0.789 against 0.814, 0.805 and 0.844 with all four, two of three seeds below by 0.03 to 0.05, a seed mean of 0.795 against 0.821; the one column still lifts the pooled cell by 0.02.

The learned layer adds nothing on top of the solve. The forced growth column with the affine beside it and no context

head scores 0.834, 0.811 and 0.809, a mean of 0.818, and the gated readout with the affine 0.810, 0.821 and 0.832, a mean of 0.821, each one of three seeds above the round-5 cell and each within its seed spread. The growth column is as good a companion to the solve as the MLP head, and no better; its gradient norm, 0.4 to 1.1 in the forced cells, says the layer trains rather than idles. Put beside rounds 2, 4 and 5, the pattern is one statement: the module's ranking is what the backbone gives it, and the physics that ranks genes on this endpoint is the solve's optimum, which the projection layer does not compute.

The embeddings do not move the ceiling. The species-LM composite scores 0.783, 0.779 and 0.773 without the affine, against ProtT5's 0.780, 0.769 and 0.773 at 2.7 times the parameters, and 0.802, 0.825 and 0.822 with it, against 0.814, 0.805 and 0.844: within noise both times. The backbone on codon frequency with the affine scores 0.747, 0.682 and 0.764, a mean of 0.731, below the pooled ProtT5 cell and 0.09 below the ladder's codon rung with FBA (0.818), so the 64 codon numbers that a support vector classifier reads well are not read well through a 12-wide transformer; the selected epochs (16, 172, 14) and the step of 0.013 in panel b say the cell never settled.

4.5 The growth column is a signed free coordinate ✗

Four rounds put the layer's growth column level with the pooled head and blind to its own k_{cat} table while one linear program per deletion lifts every model by 0.05. The probe in Table 20 reads the column itself. It rebuilds a forced cell from its result file, deletes each of the 982 labeled genes through the same batches training used, and reads v_{growth} at initialization and under the trained weights, scoring the negative column against the label (a deletion that stops growth is essential) and the column against FBA's ratio.

Table 20: The flux layer's growth column, read for every labeled deletion at initialization and under the cell's trained weights (`flux_growth_probe.py`). Median v_{growth} over the 982 labeled deletions (mmol/gDW/h; the wild type's FVA range on YPD is 0.48 to 0.54); AUROC of the NEGATIVE growth column against the released label on the train, validation and held-out genes, so 0.5 is chance and a value below 0.5 is a ranking with the sign of growth inverted; Spearman of the column against FBA's knockout-to-wild-type ratio on the shipped bounds and on YPD.

cell	weights	median v	train	val	held-out	ρ shipped	ρ YPD
`w13_fluxbio_e80_s2700`	initialization	-0.27	0.618	0.555	0.547	+0.12	+0.11
`w13_fluxbio_e80_s2700`	trained (epoch 77)	-2.41	0.091	0.266	0.241	-0.26	-0.22
`w15_bio_s0_e200_s2700`	initialization	-0.27	0.618	0.555	0.547	+0.12	+0.11
`w15_bio_s0_e200_s2700`	trained (epoch 108)	-1.43	0.048	0.258	0.228	-0.27	-0.24
`w18_biobox_e200_s2700`	initialization	45.08	0.402	0.389	0.326	-0.09	-0.05
`w18_biobox_e200_s2700`	trained (epoch 180)	0.08	0.103	0.250	0.233	-0.25	-0.22
`w18_biobox_e200_s2701`	initialization	50.26	0.512	0.503	0.603	-0.14	-0.08
`w18_biobox_e200_s2701`	trained (epoch 95)	0.02	0.637	0.696	0.609	+0.10	+0.05
`w18_biobox_e200_s2702`	initialization	53.07	0.572	0.602	0.506	+0.17	+0.15
`w18_biobox_e200_s2702`	trained (epoch 140)	0.27	0.125	0.265	0.215	-0.28	-0.23
`w18_bioprojbox_e200_s2700`	initialization	0.00	0.500	0.500	0.500	–	–
`w18_bioprojbox_e200_s2700`	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
`w18_bioprojbox_e200_s2701`	initialization	0.00	0.500	0.500	0.500	–	–
`w18_bioprojbox_e200_s2701`	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
`w18_bioprojbox_e200_s2702`	initialization	0.00	0.500	0.500	0.500	–	–
`w18_bioprojbox_e200_s2702`	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
`w21_warmbio_e200_s2700`	initialization	-0.07	0.247	0.259	0.337	-0.35	-0.43
`w21_warmbio_e200_s2700`	trained (epoch 112)	-0.10	0.272	0.215	0.383	-0.38	-0.38
`w21_warmbio_e200_s2702`	initialization	-0.10	0.304	0.305	0.392	-0.40	-0.44
`w21_warmbio_e200_s2702`	trained (epoch 157)	-0.20	0.070	0.114	0.215	-0.37	-0.39
`w21_warmgate_e200_s2700`	initialization	-0.07	0.247	0.259	0.337	-0.35	-0.43
`w21_warmgate_e200_s2700`	trained (epoch 181)	-0.10	0.422	0.328	0.410	-0.39	-0.45
`w24_warmoffbio_e200_s2700`	initialization	0.38	0.798	0.698	0.683	+0.47	+0.57
`w24_warmoffbio_e200_s2700`	trained (epoch 53)	0.55	0.689	0.638	0.623	+0.54	+0.64
`w24_warmoffgate_e200_s2700`	initialization	0.38	0.798	0.698	0.683	+0.47	+0.57
`w24_warmoffgate_e200_s2700`	trained (epoch 85)	0.55	0.735	0.748	0.670	+0.56	+0.64
`w26_lpbio_e200_s2700`	initialization	0.00	0.500	0.500	0.500	–	–
`w26_lpbio_e200_s2700`	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
`w26_lpgate_e200_s2700`	initialization	0.00	0.500	0.500	0.500	–	–
`w26_lpgate_e200_s2700`	trained (epoch 78)	0.00	0.500	0.500	0.500	–	–

At initialization the column carries a weak signal in the right direction: 0.55 on the held-out genes and a Spearman of +0.11 with the solve, nothing like FBA's 0.65, and no deletion reaches zero growth. Training inverts it. Under the round-4 forced cell's weights the negative column reads 0.23 on the held-out genes, which is that cell's 0.77 with the sign of growth reversed: the affine learned a positive slope, an essential deletion gets the *higher* growth, and the column is anticorrelated with the solve (−0.25). The round-2 forced cell does the same (0.24, −0.26). The median growth under the trained weights is −1.4 and −2.4 mmol/gDW/h against a wild-type FVA range of 0.48 to 0.54 on YPD: for most

deletions the column is negative, which growth cannot be.

The column can do this because the arm's alternations end on the balance projection, so $Sv = 0$ holds to machine precision and the box does not: 58 percent of reactions sit outside their bounds at the selected epoch (0.64 on the round-2 cell, 0.59 on the round-6 cell), the growth reaction's floor of zero among them. v_{growth} is then a signed, unbounded coordinate the backbone writes through the abundance head, and the readout uses it as a free latent. That is the mechanism behind the shuffled-k_{cat} parity and behind the layer matching the head: the quantity the forced readout reads is not growth.

4.6 Round 7: the growth column with the bounds enforced ✗

Round 7 (jobs 2257 and 2258, three cells per GPU) puts the forced readout on two layers whose bounds hold exactly: the same alternations ending on the clip instead of the balance, and the pure box form, $v = \ell + (u - \ell)\,\sigma(z)$, with no projection at all. Same cell otherwise, batch 32 with the class weight at 200 epochs, constraint scale 0, three seeds, paired against the round-4 forced cell and the pooled cell (Table 21, Figure 19); the probe runs on every cell (Table 20).

Table 21: Round 7, the growth column with the bounds enforced. The round-3 pooled cell, the round-4 forced growth column (projection ending on the balance, box open), and the forced growth column on two layers whose bounds hold exactly: the projection ending on the box, and the pure box parameterization. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	**0.777**	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
growth column only, projection ending on the box	0.500	0.000	0.500, 0.500, 0.500 ↓	0.500 [0.50, 0.50]	2, 2, 2	0.0000
growth column only, box parameterization	0.720	0.097	0.767, 0.609, 0.785	**0.781** [0.69, 0.87]	180, 95, 140	0.0074

With the clip last, the growth column is zero for every one of the 982 deletions, at initialization and after training: the clip pins the growth reaction at its floor, the cells score 0.500 exactly on three of three seeds and select epoch 2, and the balance residual, the loose end now, is 0.85 to 1.0 of the layer's own turnover. With the box form the column lives in $[0, u]$ and starts near $u/2$ (median 45 to 53 mmol/gDW/h against a wild-type 0.5); $Sv = 0$ is then never approached, the balance residual reads 2.0 at every epoch, and the cells score 0.767, 0.609 and 0.785, a seed mean of 0.720 below the pooled cell's 0.774. The probe on the trained box cells inverts the sign on two of three seeds as before (0.23 and 0.22 on the held-out genes, Spearman -0.25 and -0.28 with the solve) and reads 0.61 with $+0.10$ on the third.

The rounds close the module question at the layer rather than at the readout. Whichever constraint the layer holds exactly, the other is violated by an amount comparable to the fluxes themselves: 58 percent of reactions out of bounds when the balance is exact, a balance residual of 0.85 to 2.0 when the bounds are. The alternations never reach the polytope that FBA optimizes over, so the growth column is either a free signed coordinate, a constant zero, or an unbalanced sigmoid, and none of them is the quantity whose optimum over that polytope ranks essential genes at 0.65 on its own and lifts the transformer by 0.05. On this endpoint the physics that helps is the solve, and the layer would have to reach feasibility before its growth column could carry it.

4.7 Round 10: the layer warm-started from the deletion's own solve ✗

If the layer cannot reach the polytope from a free proposal, it can be handed a point on it. One parsimonious FBA solve per labeled gene on YPD, the medium the essentiality cell's box is built on, gives a reference flux per deletion (`ess_reference_flux_table.py`, 1,121 genes, 104 of them with zero growth, about eight minutes of CPU), and under the warm-start parameterization the layer proposes that reference plus a null-space residual of relative size α before its alternations. The growth column then starts at the deletion's own pFBA growth rather than at a free coordinate, and the negative of that growth ranks the 195 held-out genes at 0.645 on its own. Round 10 (jobs 2284 to 2287) runs the forced growth column and the gated column with the context head on the anchored layer at constraint scale 0, with α frozen at 0.01, learned from 0.01, and frozen at 0.1, each at the three seeds, paired against the round-4 forced and gated cells and the round-5 affine (Table 22, Figure 20). Two reads were fixed first: the probe at initialization must find the growth column at the solve's level, about 0.65 on the held-out genes, or the warm start did not reach the readout; and the gated cell passes if it sits at the round-5 affine's 0.82 with three of three seeds above the round-4 gated cell.

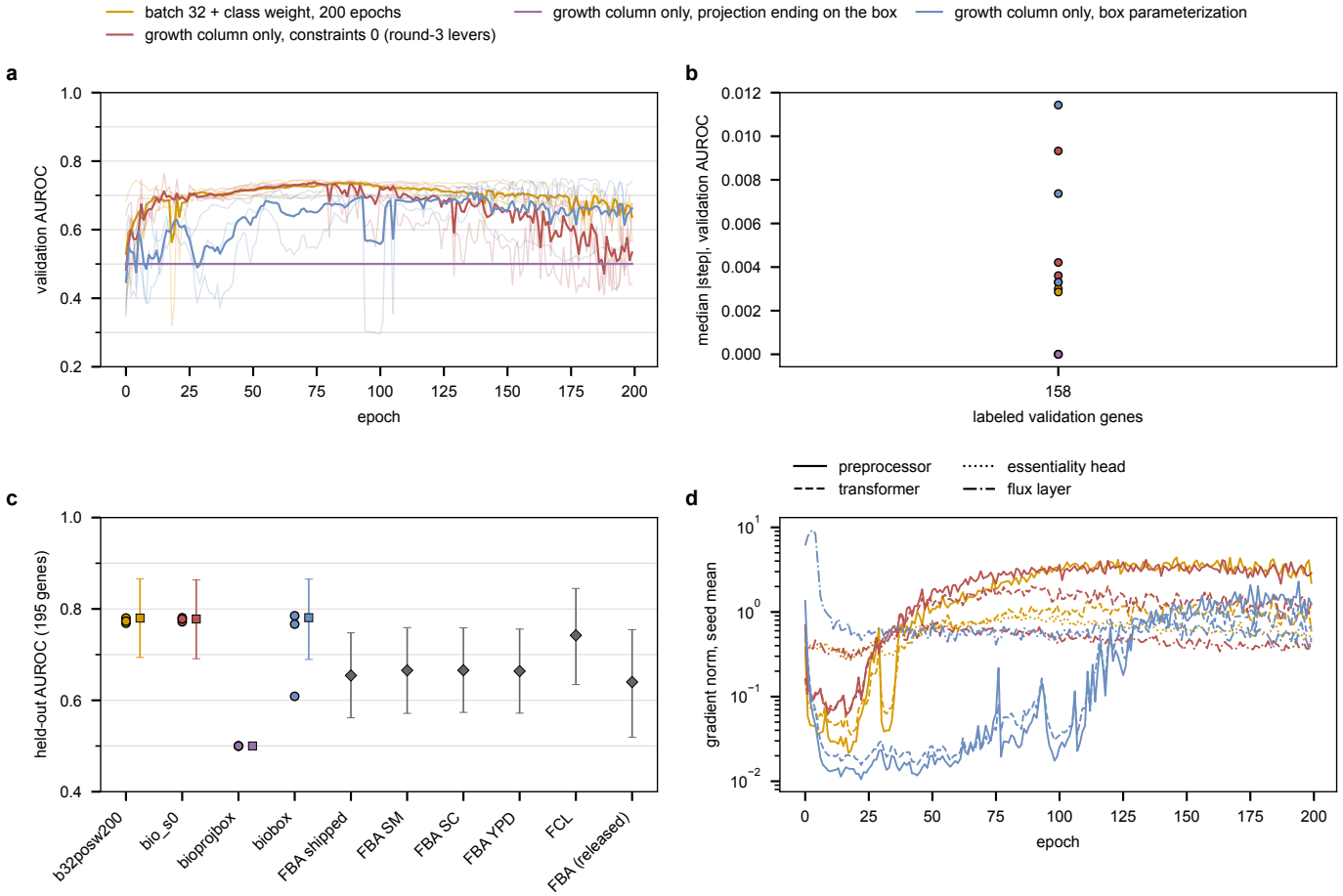


Figure 19: Round 7, the growth column with the bounds enforced. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w18`.

Table 22: Round 10, the flux layer warm-started from the deletion's own solve. The pooled cell, the round-4 forced and gated growth columns, and the round-5 FBA affine, against the forced and gated growth columns whose layer proposes the deletion's own parsimonious FBA flux on YPD plus a null-space residual of relative size alpha, with alpha frozen at 0.01, learned, or frozen at 0.1. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
growth column, pFBA warm start, alpha 0.01 frozen	0.704	0.073	0.629, 0.706, 0.776	0.801 [0.71, 0.90]	112, 144, 157	0.0222
gated column + context, pFBA warm start, alpha 0.01 frozen	0.776	0.034	0.810, 0.776, 0.742	0.784 [0.70, 0.87]	181, 70, 4	0.0037
gated column + context, pFBA warm start, alpha learned	0.775	0.035	0.812, 0.772, 0.741	0.796 [0.72, 0.87]	146, 71, 4	0.0036
gated column + context, pFBA warm start, alpha 0.1 frozen	0.751	0.044	0.784, 0.767, 0.701	0.777 [0.69, 0.86]	79, 83, 3	0.0035

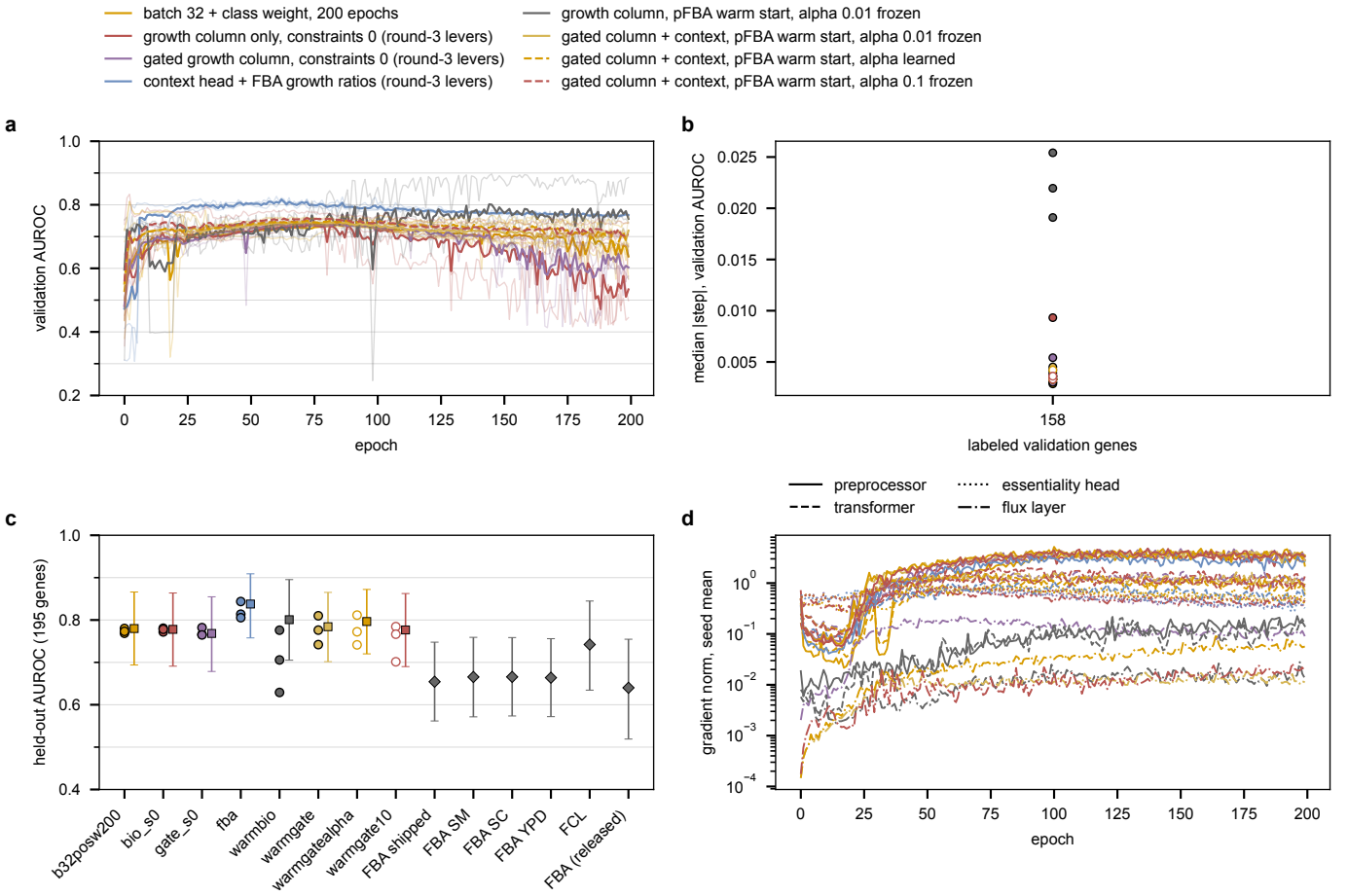


Figure 20: Round 10, the layer warm-started from the deletion's own solve. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w21`.

Neither read holds. The probe at initialization finds the growth column negative (median −0.07 to −0.10 mmol/gDW/h against references of 0 to 0.5) and anti-correlated with the solve it was started from: Spearman −0.43 to −0.44 against the YPD ratio, and the negative column ranks the held-out genes at 0.34 to 0.39, which is the solve's ranking with its sign reversed. The layer's own diagnostics say where the reference goes: the anchored layer's box is the GEM's bounds tightened by the enzyme-capacity ceilings and the second-law directions, the reference violates it on 28 to 39 percent of reactions at initialization, and the proposal the alternations return sits 0.45 to 0.57 of the reference's norm away from it at $\alpha = 0.01$. The projections that make the flux balanced move it half its length off the point it was handed and reverse the growth column's correlation with the solve on the way. Trained, the forced cells read 0.629, 0.706 and

0.776 against the round-4 forced cell's 0.772, 0.781 and 0.778, down on three of three seeds, and the probe on their trained weights inverts further (0.22 to 0.41); the gated cell reads 0.810, 0.776 and 0.742 against 0.766, 0.782 and 0.765, one seed up and two down, with seed 2702 selecting epoch 4 and its gate ending at −2.0. Letting α learn changes nothing on the held-out genes (0.812, 0.772, 0.741) while α grows to 0.19, 0.02 and 0.11 and the box violation on one seed rises from 0.30 to 0.45, the job-1946 failure mode in miniature; α frozen at 0.1 reads 0.751. The warm start hands the layer the solve and the anchored layer's constraint set does not keep it. Round 13 repeats the warm start on the stoichiometry-only layer, whose box is the GEM's own bounds, so the reference is inside it by construction.

4.8 Round 13: the warm start on the stoichiometry-only layer ✗

Round 13 (jobs 2297 and 2298) repeats the warm start on the stoichiometry-only layer, whose box is the GEM's own bounds scaled by the availability gate, with no capacity ceiling and no second-law direction, forced and gated, α frozen at 0.01, three seeds each (Table 23, Figure 21). The reads are round 10's: the probe at initialization at the solve's level, and the gated cell at the round-5 affine's 0.82 with three of three seeds above the round-4 gated cell.

Table 23: Round 13, the warm start on the stoichiometry-only layer. The pooled cell, the round-4 forced and gated columns and the round-10 warm-started columns on the anchored layer, against the same warm start on the stoichiometry-only layer, whose box is the GEM's own bounds rather than the capacity- and second-law-tightened one, so the deletion's pFBA reference sits inside it. Alpha frozen at 0.01, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
growth column, pFBA warm start, alpha 0.01 frozen	0.704	0.073	0.629, 0.706, 0.776	0.801 [0.71, 0.90]	112, 144, 157	0.0222
gated column + context, pFBA warm start, alpha 0.01 frozen	0.776	0.034	0.810, 0.776, 0.742	0.784 [0.70, 0.87]	181, 70, 4	0.0037
growth column, GEM box only, pFBA warm start, alpha 0.01 frozen	0.664	0.058	0.654, 0.726, 0.612 ↓	0.676 [0.55, 0.79]	53, 2, 23	0.0236
gated column + context, GEM box only, pFBA warm start, alpha 0.01 frozen	**0.816**	0.019	0.794, 0.829, 0.825 ↑	**0.802** [0.72, 0.88]	85, 5, 118	0.0042

Both reads hold, and this is the first module cell in the document that carries the solve. At initialization the growth column is positive (median 0.38 mmol/gDW/h), its Spearman with the solve is +0.47 to +0.57 across the four media, and its negative ranks the held-out genes at 0.683, above FBA's own 0.654 to 0.666 on the same genes; the alternations leave the reference where it was, 0.3 percent of its norm away, with a box violation of 0.24 to 0.29 that is the availability gate's untrained scaling of the bounds rather than a displacement of the flux. The forced cells read 0.654, 0.726 and 0.612, a mean of 0.664, which is the solve's number: trained, the column keeps its sign (Spearman +0.54 to +0.64) and its ranking (0.62 to 0.67), and the layer's learned parts add nothing to it. The gated cells read 0.794, 0.829 and 0.825 against the round-4 gated cell's 0.766, 0.782 and 0.765, up by 0.028, 0.047 and 0.060 on three of three seeds, a seed mean of 0.816 against the affine's 0.821, and on the two seeds that train past epoch 5 the gate ends at +0.97 and +0.90, the first positive gates in the document: the head adds the growth column with its physical sign. The third seed selects epoch 5 with its gate at zero and still reads 0.829, the seed-to-seed spread the round-9 cells also show.

Put beside round 10, the comparison isolates the loss. The same reference on the anchored layer arrived sign-reversed and half its norm away; on the GEM's own box it arrives intact. The capacity ceilings and the second-law directions, tightened onto a flux that was solved without them, are what the projections could not reconcile, and the layer that solves (round 15) applies those constraints inside the solve instead of after it.

4.9 Round 15: the layer that solves ✗

The projection route carries the solve only when handed it. Round 15 replaces the alternations with the solve: under the `lp` parameterization the layer's flux is the growth optimum of the linear program $\max\{v_{\text{growth}} : Sv = 0,\ \ell \le v \le u\}$ over the layer's own bounds, one HiGHS solve per genotype in a forked pool, and the backbone reaches the prediction through the bounds alone, with the gradient the bound duals give (`torchcell.metabolism.lp_layer`; the envelope theorem, checked against finite differences on the GEM). On the anchored layer the bounds are the GEM's scaled by availability and capped by the enzyme-capacity ceilings from the abundance head, so this is the GECKO construction with a learned enzyme pool. Round 15 (jobs 2304 to 2306) runs it forced, forced with the second-law box, and gated, three seeds each (Table 24, Figure 22).

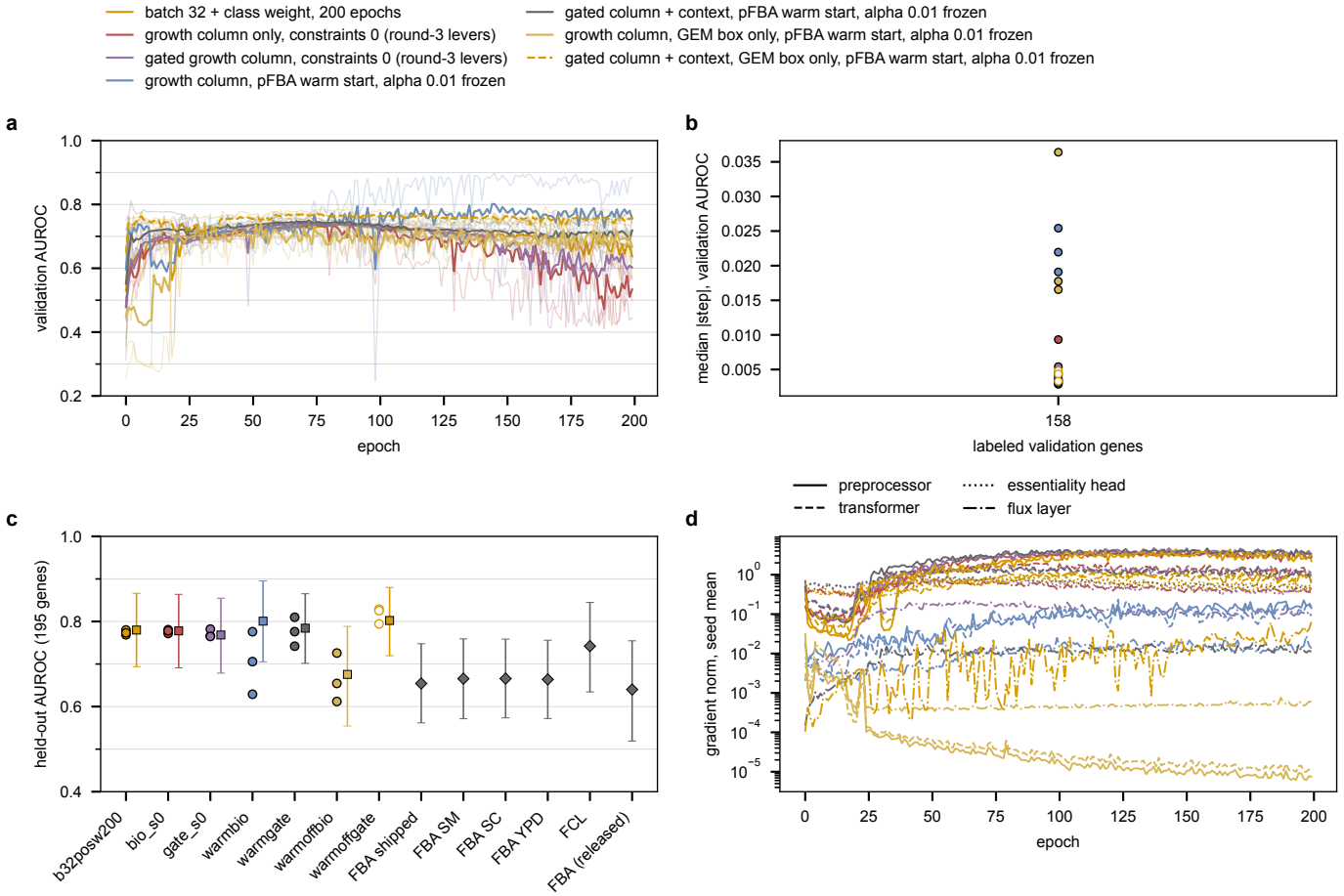


Figure 21: Round 13, the warm start on the stoichiometry-only layer. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w24`.

Table 24: Round 15, the layer that solves. The pooled cell, the round-4 forced and gated growth columns and the round-5 FBA affine, against a flux layer whose flux is the growth optimum of a linear program under the layer's own bounds (availability and the enzyme-capacity ceiling from the abundance head), with the bound duals as its gradient: the forced growth column, the same with the box tightened to the second-law direction before the solve, and the gated column with the context head. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
growth column, the layer that solves	0.500	0.000	0.500, 0.500, 0.500 ↓	0.500 [0.50, 0.50]	2, 2, 2	0.0000
growth column, the layer that solves, second-law box	0.500	0.000	0.500, 0.500, 0.500 ↓	0.500 [0.50, 0.50]	2, 2, 2	0.0000
gated column + context, the layer that solves	0.773	0.012	0.771, 0.786, 0.763	0.770 [0.68, 0.86]	78, 81, 63	0.0028

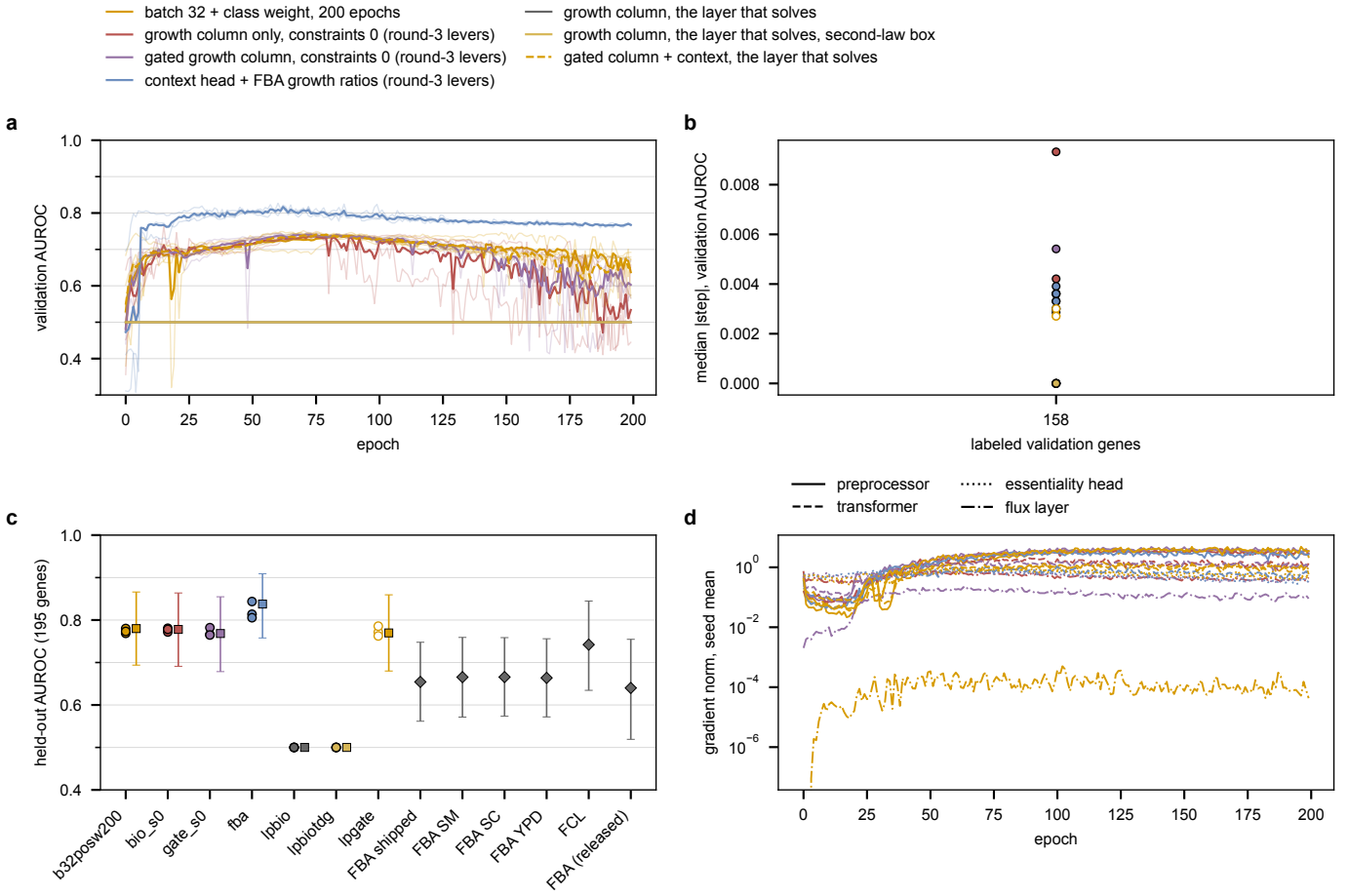


Figure 22: Round 15, the layer that solves on the anchored box. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w26`.

Every forced cell scores 0.500 exactly and selects epoch 2, because the program is infeasible for every genotype at every epoch: the probe finds the growth column at zero for all 982 deletions at initialization and under the trained weights. The gated cells read 0.771, 0.786 and 0.763, the pooled cell's 0.774, with gates within 0.07 of zero, which is the head working alone beside a dead path. The cause is measured on one genotype's bounds taken from the layer: under the GEM's own bounds the program solves at the wild-type growth of 0.0858, and under the layer's bounds it has no feasible point, with or without the maintenance flux released and with the deleted gene's reactions reopened. What differs is the capacity ceiling: the layer's upper bounds have a median of 0.084 mmol/gDW/h and a tenth of the reactions below 0.007, against 100 on the GEM, because the predicted k_{cat} tier multiplied by the abundance head's nominal abundances gives ceilings two to four orders below the fluxes growth needs. The alternating layer never noticed,

because it never satisfied the box; the solve cannot proceed without it. That is the same fact as round 10's, read exactly: the enzyme-capacity box as parameterized contains no growing flux, and every arm built on it has been reading a flux off the polytope. Round 15b runs the solve on the stoichiometry-only box, where round 13 showed the polytope holds growth, and a capacity ceiling that admits growth (the pool scaled so the wild type is feasible, GECKO's saturation calibration) is the module's next construction.

4.10 Where the gradient goes ✗

Every cell logs a gradient norm per module each epoch, and read across the module arms (Figure 23) they say which of the three failure modes each arm is in. On the forced arms the whole gradient passes through the layer and the layer passes it on: the flux layer carries 10 to 40 percent of the summed module norm and the preprocessor's norm climbs to 1 to 3, the same level as under the pooled head, so the backbone is trained through the layer and the growth column is a trained, free coordinate that ranks no better than the head. On the gated arms the layer carries 0.3 to 3 percent, the head carries the rest, and the gate ends between -0.2 and $+0.2$ across seeds under constraint scale 0 (seed mean $+0.05$) and at $+0.2$ on every seed with the FBA affine beside it, so the backbone is trained through the MLP and the flux path is a small correction. On the warm-started forced arm with α frozen the backbone receives no gradient at all: the preprocessor's norm sits at 0.02 to 0.1 for the whole run, two orders below every other arm, because the flux is 99 percent a fixed reference and only the two parameters of the growth affine train; the cell is the affine on the projected reference, and the projection reversed its sign. The warm-started gated arms leave the flux path at 0.3 percent of the gradient and push the gate to -0.8, which is the head learning to subtract a sign-reversed growth column. None of the arms is gradient-starved except the one that was meant to be a fixed function, so the module's null on this endpoint is not a gradient-flow problem; the gradient reaches the layer, and what the layer computes is not growth.

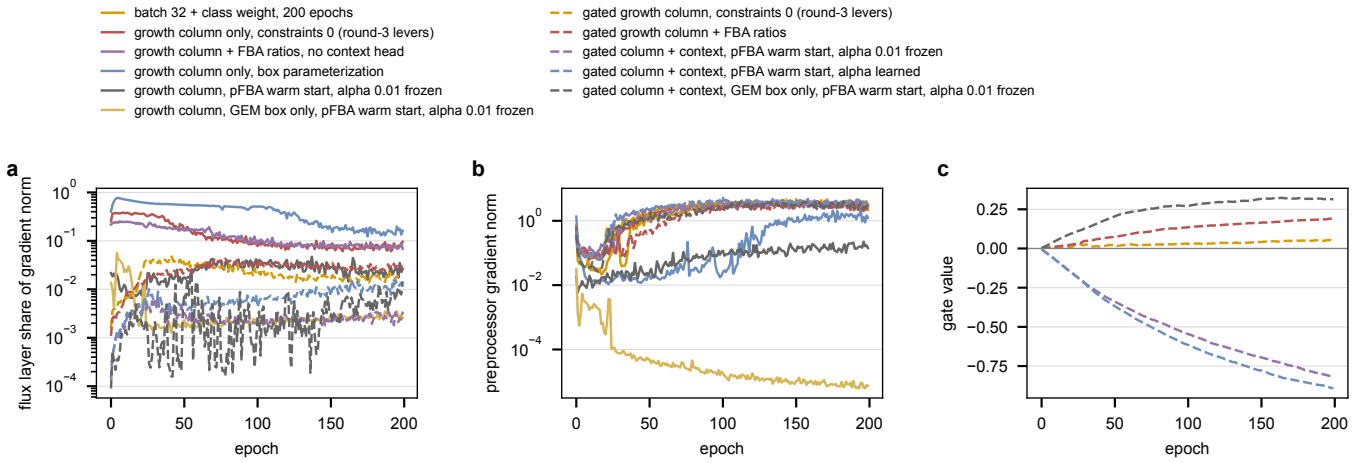


Figure 23: Where the gradient goes on the module arms. a, The flux layer's share of the summed per-module gradient norm per epoch, seed mean, on every module arm of rounds 4 to 13; solid lines are the forced arms and the pooled reference, dashed the gated arms. **b**, The preprocessor's gradient norm per epoch, seed mean, on the same arms. **c**, The gate's value per epoch on the gated arms. Written by `experiments/028-gene-essentiality/scripts/grad_flow_readout.py`.

4.11 Why the solve helps and the layer does not ✗

What FBA contributes to the prediction is one number per deletion: the maximum of growth over the polytope $\{Sv = 0,\ \ell \le v \le u\}$ with the deleted gene's reactions closed. That number ranks the held-out genes at 0.65 alone, its four-medium version lifts a logistic regression, a random forest and the transformer by 0.05 each, and the solve has more to give than the number: eight statistics of the gene's reactions under the wild-type solve reach 0.80 with no sequence, and the whole deletion flux vector 0.77. The value is in the optimum over the constraint set and in the flux that attains it.

The module holds the same constraint set and computes something else. Its flux is K alternating projections of a proposal the backbone writes, and the rounds measure, in order, that the projections stop short of the polytope (58 percent of reactions out of bounds when the balance is exact, a balance residual of 0.85 to 2.0 when the bounds are), that the growth column of a flux off the polytope is a free coordinate the readout learns to use with either sign (the probe: 0.23 on the held-out genes under trained weights, Spearman -0.25 with the solve), that the learned parts of the layer are not what the column carries (the shuffled-k_{cat} control matches the real table), and that the layer's constraint set rejects the solve when the solve is supplied as the proposal (round 10: the reference violates the anchored box on a

third of its reactions and the returned flux sits half the reference's norm away, sign reversed). The gradient is not the cause; it reaches the layer on every arm that lets it.

The comparison also bounds what a feasible layer could add. A layer that reached the polytope would return a feasible flux, not the optimal one, unless the backbone learned to write proposals whose projection is the optimum; the solve finds the optimum by construction. And a growth column is one number, where the solve's flux vector scores 0.11 to 0.13 more AUROC on the ladder than its growth ratio does. A module that helped on this endpoint would therefore have to solve rather than project, with the backbone setting the bounds it solves under (the capacity ceilings from the abundance head are that interface already), and read more of the resulting flux than its growth column. The first of those is a linear program per genotype per forward pass with the gradient taken through its dual, about 50 milliseconds each; it is untested here and is the one construction the rounds have not ruled out. Round 13, the warm start on the stoichiometry-only layer, is the last test of the projection route: its box is the GEM's, so the reference sits inside it, and if the growth column still leaves the solve's ranking there, the alternations themselves are what moves it.

5 Open ✗

The held-out number is close to saturated: the round-12 ensemble's interval, 0.893 [0.84, 0.94], spans what 195 genes with 31 essential can resolve, and every lever on the transformer side has been moved (width, depth, the preprocessor, the budget, the class weight, six encodings). What remains is attribution and the module.

- The pathway multi-hot carries the round-9 gain (round 14); which pathways, and whether a pathway-level flux statistic from the solve would substitute for the annotation, is not measured.
- The round-16 cell (the round-11 affine on the round-9 backbone) and the round-9 cell are one cell to the resolution of 195 genes (rounds 16 and 17); a larger held-out set, such as the genome-wide labels scored on genes the released split never touched, is what would separate them.
- Round 13: the pFBA warm start on the stoichiometry-only layer. If its growth column reads at the solve's level at initialization and keeps it, the projection route carries the solve and the anchored constraint set was the loss; if not, the alternations are.
- A layer that solves. The one module construction the rounds have not ruled out is a linear program per genotype inside the forward pass, with the backbone's capacity ceilings as its bounds and the gradient through the LP's dual prices, reading the optimal flux rather than a growth column. One solve costs about 50 milliseconds, so a 200-epoch cell at batch 32 is about two hours of CPU per seed. Untested.
- The module's own hyperparameters have not been moved on this endpoint: the k_{cat} tier (measured against predicted), the thermodynamic source, the number of alternations. Given the probe and round 10, none of them reaches the polytope.